A Novel Nonlocal-Aware Pyramid and Multiscale Multitask Refinement Detector for Object Detection in Remote Sensing Images

Zhanchao Huang, Wei Li, *Senior Member, IEEE*, Xiang-Gen Xia, *Fellow, IEEE*, Xin Wu, *Member, IEEE*, Zhaoquan Cai, and Ran Tao, *Senior Member, IEEE*

***Abstract*— Object detection (OD) is an important task of computer vision and has been widely used in many fields, including remote sensing (RS). However, the complex scenes, large-scale variation, and dense instances of RS bring huge challenges to OD. To meet these challenges, a novel Nonlocal-aware Pyramid and Multiscale Multitask Refinement Detector (NPMMR-Det) is proposed. Specifically, nonlocal-aware pyramid attention (NP-Attention) is designed for guiding a neural network model to focus more on efficient features and suppress background noise. Then a multiscale refinement feature pyramid network (MSR-FPN) is proposed to fuse the multiscale context features extracted by the NP-Attention guided neural network and adjust the optimal receptive field. In order to use these features more effectively, a multitask refinement head called MTR-Head, with offset sharing and a modulation mechanism, is developed to refine the feature misalignment between the localization task and the classification task. Extensive experiments performed on two public RS data sets demonstrate that the proposed NPMMR-Det achieves competitive performance compared with state-of-the-art methods.**

***Index Terms*— Attention, multiscale, multitask, object detection (OD), remote sensing (RS) images.**

I. Introduction

As A fundamental task of computer vision image analysis, object detection (OD) is widely used in many fields. The pipeline of the classic OD method is mainly based on the Deformable Part Models (DPM) [1], which has also been applied to the remote sensing (RS) field by Cheng *et al.* [2] and Chen *et al.* [3] to detect vehicles in aerial images. The DPM is a detection method using handcrafted features and sliding windows but is limited by its complexity of feature design and the inefficiency of object search.

With the rapid development of deep learning (DL) technology, OD methods based on convolutional neural networks (CNN) continue to emerge [4]. They can be divided broadly into two-stage detectors like Faster R-CNN [5], R-FCN [6], and Cascade R-CNN [7], and one-stage detectors like YOLO [8], SSD [9], and RetinaNet [10]. A two-stage detector focuses on detection accuracy but has a slower computational speed, while a one-stage detector pursues the trade-off of accuracy and speed. With more recent research developments, the boundaries between these two approaches get harder to tell. Benefiting from the availability of a large number of RS image resources, many excellent detectors have been successively developed, such as the two-stage detectors RI-CNN [11], RIFD-CNN [12], FADet [13], SCRDet [14], CAD-Net [15] and the one-stage detectors SBL [16], FMSSD [17], and EFR [18].

Detecting objects in RS images involves special challenges compared to detecting objects in general natural images. This is because the RS images are characterized by complex scenes, large scale variation, and dense instances, which lead to misalignments between features extracted by the OD model and real objects. Three aspects of these misalignments and problems in the existing RS detectors are summarized below.

First, RS images are usually more complicated than general images. They often contain complex background interference, which leads to misalignments between the network's focus and the real objects in the images. For this reason, a visual attention mechanism has been widely used in remote sensing object detection (RSOD) [13]–[15], [19] to highlight the features from an object area and suppress the background noise. However, the visual attention methods used in these detectors simply reweight the feature maps, and the adjustment of these weight masks depends on the local convolution features [20]. Because of the characteristics of RS images, a better attention mechanism for OD tasks should learn the multiscale nonlocal dependencies and use the edge features that are significant for objects. In addition, it should be efficient and plug-and-play.

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Second, a distinctive characteristic of RS images is that the objects have a large scale-variation, much greater than that for natural images, which leads to misalignments between the scale of the features extracted and the scale of the real objects. The feature pyramid [21] has been commonly used to deal with scale variation, but a single pyramid with only a top-down branch does not significantly improve the detection performance of large-scale objects while the efficiency of a double-pyramid structure like FFA [22] is low. Moreover, whether the receptive field of the convolutional layer matches the object's scale is also very important. The FMSSD [17] has done some work in this regard based on the ideas of ASPP [23]. In view of these problems, a better multiscale pyramid should extract and fuse multiscale context features more efficiently, and receptive fields of convolutional layers should match the corresponding scales of the objects.

Third, RS images contain large numbers of densely arrayed objects, which results in feature misalignment between the localization subtask that focuses on the transformation equality of features and the classification subtask that focuses on the transformation invariance of features in OD [24]. These features are more likely to interfere with each other in complex RS scenes. For this reason, a multitask detection head with interpretability is needed to refine this misalignment and secure better performance.

In summary, the misalignment challenges related to these aspects of RS images, i.e., the network's focus and the real objects, the scale of features and the scale of objects, and the localization subtask and classification subtask, may limit the performance of detectors. To deal with these issues, this article proposes a novel nonlocal-aware pyramid and multiscale multitask refinement detector (NPMMR-Det). The main contributions of this article are summarized as follows.

1) The nonlocal-aware pyramid attention (NP-Attention) is designed. It adopts pyramid pooling to capture the multiscale long-distance dependencies on the feature maps and employs an edge enhancement mechanism to guide the attention module focusing on multidirectional edge features adaptively. NP-Attention solves the misalignment between the network's emphasis and the real objects in the images efficiently with high performance.
2) The multiscale refinement feature pyramid network (MSR-FPN) is presented. MSR-FPN adopts a new feature pyramid structure that merges the top-down and bottom-up branches, and improves the convolution sets through a novel dilated convolutional parallel structure with multiple receptive fields, in order to refine the misalignment between the scale of features and the scale of objects.
3) The multitask refinement head (MTR-Head) with offset sharing and modulation mechanism is developed, and a modified loss function with the scale adaptive factor is designed to alleviate feature misalignment between the localization and the classification.

This article is organized as follows. Section II reviews related works on the visual attention mechanism, multiscale detection, and detection head. Section III explains the proposed NPMMR-Det in detail. Section IV presents the experimental results and discussion. Section V concludes and identifies potential future work.

II. Related Works

A. Visual Attention Mechanism

As a common approach to highlight potential object areas while suppressing unrelated areas in feature maps, the visual attention mechanism has been widely employed in the CNN model [25]. This mechanism is mainly used to solve the problem of the misalignment between the network's emphasis and the real objects in the images. It is essentially a reweighting operation on different dimensions (such as the channel dimension [25], the spatial dimension [20], or both [26], [27]) of the feature maps. The weights can be adjusted adaptively during CNN model training.

In the field of RSOD, various attention methods have also been introduced, such as the FADet [13], the MDA-Net in SCRDet [14], and the SAM block in CAD-Net [15]. However, these methods are simple attention methods in which the adjustment of the masks completely depends on the network training, and the necessary long-distance dependence between spatial features is not constructed [20]. Because the receptive fields of convolutional kernels are local, the parameter adjustment of the aforementioned masks is only related to the local pixels. This is one of the reasons why their performance is unstable and does not offer much improvement over earlier works.

The literature [20], [27], [28] has suggested that local-local and local-global dependencies are very important for computer vision tasks like classification, detection, and segmentation. For learning nonlocal dependencies, self-attention is introduced into the visual attention mechanism by the Nonlocal Network [20]. The CGNL [29] introduced channel attention and improves efficiency based on the Nonlocal Network. Cao *et al.* [27] combined nonlocal attention with SENet and proposed the Global Context Network (GCNet) with a low computational cost of global context modeling, which achieved better results on various computer vision tasks. But why the nonlocal block is not position dependent when training is still not clear. The Dual Attention Network (DANet) proposed by Fu *et al.* [30] captured global feature dependence in both the spatial dimension and channel dimension. The Criss-Cross Network (CCNet) [31] designed a criss-cross module to obtain context information on its criss-cross path and to capture long dependencies through a recurrent operation. Although the attention model proposed by Zhu *et al.* [32] introduced the patch-patch spatial dependence based on the Nonlocal Network, it did not learn the dependence on the channel dimension. Moreover, these methods mentioned above do not effectively use the top-view images' features when they are applied in RSOD.

B. Multiscale Detection

The problem of scale variation in objects is critical for the detection task, especially in the detection of RS objects. The misalignment between the feature scales of the neural network and the object scales affects the accuracy of OD.

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For solving the problem of scale variation, a multiscale image pyramid is an intuitively appealing idea. For example, OverFeat [33] applied a convolutional network as a sliding window detector on a multiscale image pyramid; SNIP [34] trained the detectors on the scale normalized image pyramids, selectively back-propagating the gradients of objects with different sizes as a function of the scale. But these methods using image pyramids are complicated and it is difficult to use them to deal with the multicategory OD tasks with varying scales.

Constructing feature pyramids in neural networks is a more efficient and widely used way to deal with the scale variation problem. SSD [9], MS-CNN [35], and many other methods [36], [37] have adopted the idea of detecting the objects with different scales on the different layers of neural networks (with different resolutions) independently. However, an object may have joint features on different scales, and the independent detection idea may result in the loss of some features.

The feature pyramid network (FPN) [21] fused a series of feature maps with different scales by constructing a multiple U-shape module between different layers of the neural network. It gets better performance and is widely exploited by state-of-the-art detectors. Recently, many studies [38]–[40] have suggested that when improving the detection accuracy of the small object, the detection accuracy of a large object is not noticeably improved. This is because the top-down pyramid structure of FPN only propagates the high-level semantic features from the deep layers of the network to the shallow layers, while not propagating the low-level features from the shallow layers of the network to the deep layers. In response to this problem, PANet [38] and FFA [22] were designed by adding a new bottom-up pyramid to the top-down pyramid. Although such a double-pyramid structure improves the multiscale OD accuracy, the calculation is also expensive. Furthermore, whether the receptive field of each branch of these feature pyramid structures is matched with the object scale also affects detection accuracy.

Some additional ideas have been proposed for scale variation in the literature. EfficientDet [39] proposed the BiFPN, which uses a neural architecture search. From the perspective of the receptive field, TridentNet [41] used a parallel branch structure with different dilation rates to detect multiscale objects.

C. Detection Head

The OD task usually includes two subtasks: object localization and classification. The localization task not only predicts the coordinates of the bounding box but also predicts the confidence that the object exists at this location. Therefore, it is very important to design a good detection head for the corresponding task.

Light-Head RCNN [42] designed an efficient head using a thin feature map and a cheap R-CNN subnet consisting of pooling and a single fully connected layer. A series of cascaded detection heads were used by the Cascade RCNN [7] and its improved methods [43]. Although such multistage detection heads improve the accuracy of detection, the efficiency is low. IoU-Net [44] introduced an extra branch to predict IoU (Intersection over Union) scores for alleviating the problem of misaligned classification and localization accuracy. Zhang and Wang [45] suggested that there is a feature misalignment between the classification and localization tasks. Based on this research, Chen *et al.* [23] proposed that these two tasks to be split into different heads, including a convolution head for bounding box regression and a fully connected head fusing the location features for classification. Furthermore, RepPoints [46] was designed to employ deformable convolution to design a detection head for learning the location offsets.

Corresponding to the detection head, many loss functions have been proposed to measure the error between the prediction of the detection head and the ground truth. He *et al.* [47] developed a bounding box regression loss using Kullback–Leibler divergence to learn the uncertainties of bounding box prediction. Rezatofighi-*et al.* [48] and Zheng *et al.* [49] designed the Generalized IoU (GIoU) loss and Distance-IoU (DIoU) loss, respectively, for the localization task.

III. Proposed NPMMR-DET Detection Framework

In response to the three misalignment problems that arise in existing detectors in the RSOD task, the NPMMR-Det, consisting of NP-Attention, MSR-FPN, and MTR-Head, is proposed. The structure of the proposed NPMMR-Det framework is shown in Fig. 1. Specifically, the proposed NP-Attention guides the backbone CNN to extract the effective features of objects and suppress background noise. Then, features with different resolutions and containing different semantic information are output to MSR-FPN from different convolutional layers of the network. The MSR-FPN constructs a novel feature pyramid structure to fuse these multiscale context features and adjusts the optimal receptive field in different branches. After that, the MTR-Heads refine the feature misalignment of different subtasks by applying the offsets and modulation mechanism; they then adjust prediction results. The blue, red, and yellow parts in Fig. 1 indicate the three respective improvements.

A. Proposed NP-Attention

As shown in Fig. 2(a), the simple attention methods [13], [15], [26] only learn the local attention features by the convolutions with the local receptive fields. The parameter adjustments of these attention methods are only related to the local pixels, for which the learning process is a complete black box with unstable performance. The nonlocal attention [20] learns the nonlocal dependencies of features by calculating the spatial correlation matrix between each pixel position and other pixel positions in the feature maps, the process of which is shown in Fig. 2(b). But this nonlocal dependence is pixel–pixel, which may cause a lot of fine noise on feature maps.

Unlike the pixel-level tasks, such as semantic segmentation or image reconstruction, OD is more concerned with nonlocal dependencies of the multiscale patch-patch (pixels in a certain area), as shown in Fig. 2(c). Moreover, the dependencies of

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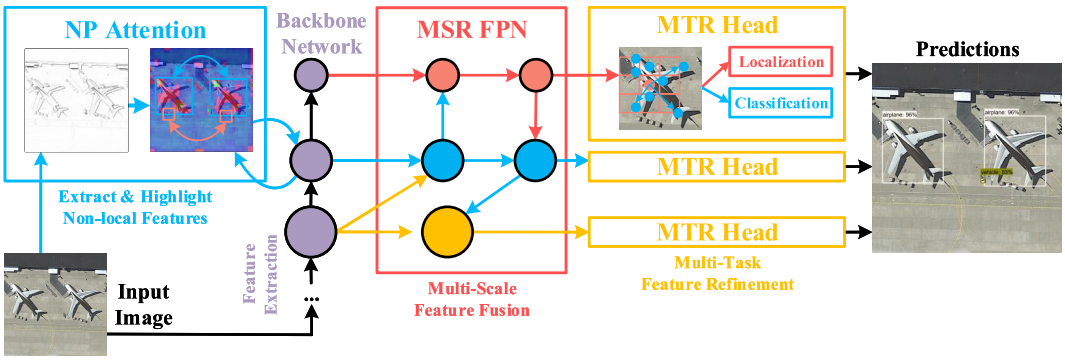

Fig. 1. Proposed NPMMR-Det Framework.

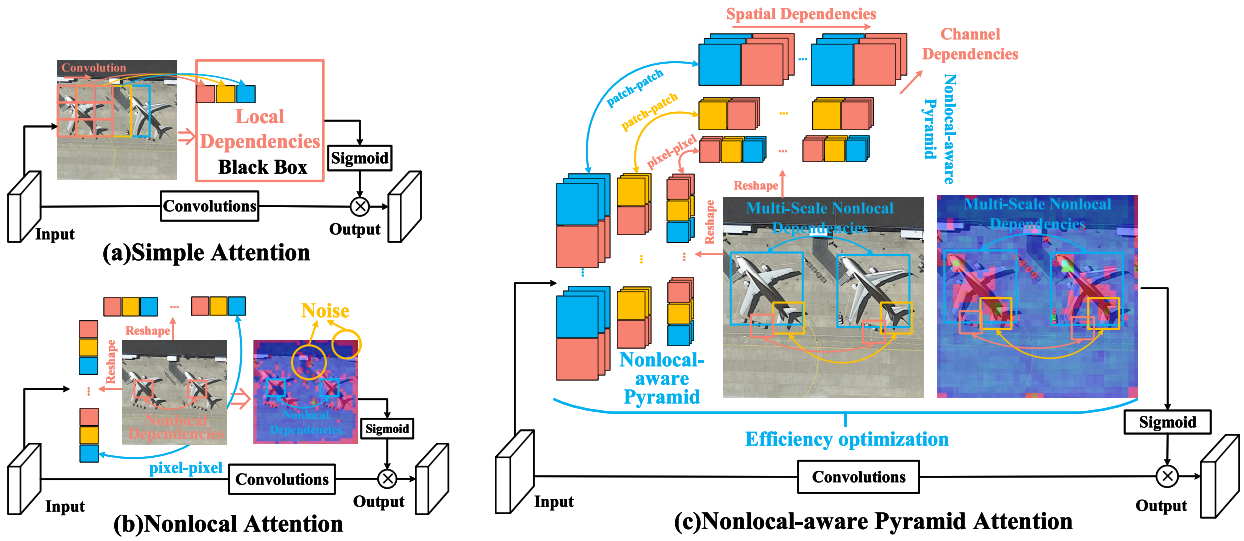

Fig. 2. Comparison of different attention models, which illustrates the motivation of proposed NP-Attention. (a) Simple Attention. (b) Nonlocal Attention. (c) NP-Attention.

channel-channel are also very important. In response to these, the motivation of NP-Attention is to learn spatial multiscale nonlocal dependencies and channel dependencies in order to refine the misalignment between the network's emphasis and the real objects in the images.

1) Nonlocal-Aware Pyramid: The features of multiscale patches are extracted and fused by multiscale max-pooling, and these feature maps are concatenated and vectorized as the input of the nonlocal attention module. The vectorized multiscale feature maps $\mathrm{vec}(\boldsymbol{X}_{NP}\boldsymbol{W})$ can be expressed as

$$\mathrm{vec}(\boldsymbol{X}_{NP}\boldsymbol{W}) = [\mathrm{vec}(\boldsymbol{X}_1\boldsymbol{W}), \mathrm{vec}(\boldsymbol{X}_3\boldsymbol{W}), \mathrm{vec}(\boldsymbol{X}_5\boldsymbol{W})] \quad (1)$$

where $\mathrm{vec}(\cdot)$ is the vectorization function that fuses the spatial dependence and channel dependence of the feature maps in the attention module [29]. $\boldsymbol{X}_{NP} \in \mathbb{R}^{3NC\times 1}$ is the nonlocal-aware pyramid feature maps, where N indicates the number of pixels in the spatial dimension ($N = rows \times cols$) and C indicates the number of channels. $\boldsymbol{W} \in \mathbb{R}^{C\times C}$ is the weight matrix

$$\boldsymbol{X}_n = \begin{bmatrix} MP\left(\boldsymbol{pat}_n^{(1,1)}\right) & \cdots & MP\left(\boldsymbol{pat}_n^{(1,\mathrm{cols})}\right) \\ \vdots & \ddots & \vdots \\ MP\left(\boldsymbol{pat}_n^{(\mathrm{rows},1)}\right) & \cdots & MP\left(\boldsymbol{pat}_n^{(\mathrm{rows},\ \mathrm{cols})}\right) \end{bmatrix} \quad (2)$$

is the pyramid feature map with scale n. In 2, $\boldsymbol{pat}_n^{(i,j)} \in \mathbb{R}^{n\times n}$ denotes the image patch, of which the coordinates of top-left are (i, j) and the size is $n \times n$. $MP(\cdot)$ denotes the operation of max-pooling with stride $= 1$.

After max-pooling, each pixel of the feature maps fuses the feature information of the patch centered on that pixel. In this way, the size of the feature map is not changed, and the correlation calculation between the multiscale patches is still transformed into the correlation calculation between the pixels as the original nonlocal attention method. The patch size is set to be 1×1, 3×3, and 5×5, which is determined by statistically analyzing the size of instances in the training images and scaling it to the size of the NP-attention feature maps. Thus, the input feature $\mathrm{vec}(\boldsymbol{X}_{NP}\boldsymbol{W}) \in \mathbb{R}^{3NC\times 1}$, and the weight is shared in multiscale feature maps.

Therefore, the three input functions (Query, Key, and Value) of NP-Attention are stated as

$$\begin{aligned} \text{Query}: \ \theta(\boldsymbol{X})_{\mathrm{vec}} &= \mathrm{vec}(\boldsymbol{X}\boldsymbol{W}_\theta) \in \mathbb{R}^{NC\times 1} \\ \text{Key}: \ \varphi(\boldsymbol{X}_{NP})_{\mathrm{vec}} &= \mathrm{vec}\left(\boldsymbol{X}_{NP}\boldsymbol{W}_\varphi\right) \in \mathbb{R}^{3NC\times 1} \\ \text{Value}: \ g(\boldsymbol{X}_{NP})_{\mathrm{vec}} &= \mathrm{vec}\left(\boldsymbol{X}_{NP}\boldsymbol{W}_g\right) \in \mathbb{R}^{3NC\times 1} \end{aligned} \quad (3)$$

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where $\theta(\cdot)$, $\varphi(\cdot)$, and $g(\cdot)$ are learnable transformations on the input. They are implemented using convolutions in neural networks. The feature vectors above contain the nonlocal multiscale dependencies between any positions of any channels. The original nonlocal attention [20] is presented as

$$\boldsymbol{Y} = f(\theta(\boldsymbol{X}), \varphi(\boldsymbol{X}))g(\boldsymbol{X}) \tag{4}$$

where correlation function $f(\theta(\boldsymbol{X}), \varphi(\boldsymbol{X}))$ computes the affinity between all positions of $\theta(\boldsymbol{X})$ and $\varphi(\boldsymbol{X})$, is transformed into

$$\mathrm{vec}(\boldsymbol{Y}) = f\big(\mathrm{vec}(\boldsymbol{X}\boldsymbol{W}_\theta), \mathrm{vec}\big(\boldsymbol{X}_{NP}\boldsymbol{W}_\varphi\big)\big)\mathrm{vec}\big(\boldsymbol{X}_{NP}\boldsymbol{W}_g\big) \tag{5}$$

where $\boldsymbol{Y} \in \mathbb{R}^{N\times C}$ denotes the output of the attention module.

2) Efficiency Optimization and Complexity Analysis: For the high complexity $O((3NC)^2)$ of 5, the Taylor series is used to approximate the $f(\cdot)$ $(P \ll NC)$ [29] as

$$\begin{aligned} &f\big(\mathrm{vec}(\boldsymbol{X}\boldsymbol{W}_\theta), \mathrm{vec}\big(\boldsymbol{X}_{NP}\boldsymbol{W}_\varphi\big)\big) \\ &\quad\approx \sum_{p=0}^{P}\left(\frac{f_\theta^{(p)}\big(\mathrm{vec}(\boldsymbol{X}\boldsymbol{W}_\theta), \mathrm{vec}\big(\boldsymbol{X}_{NP}\boldsymbol{W}_\varphi\big)\big)}{p!}\right. \\ &\qquad\left.\times\ \ (\mathrm{vec}(\boldsymbol{X}\boldsymbol{W}_\theta) - \mathrm{vec}(\boldsymbol{X}\boldsymbol{W}_\theta)_0)^p\right) \\ &\qquad+\sum_{p=0}^{P}\left(\frac{f_\varphi^{(p)}\big(\mathrm{vec}(\boldsymbol{X}\boldsymbol{W}_\theta), \mathrm{vec}\big(\boldsymbol{X}_{NP}\boldsymbol{W}_\varphi\big)\big)}{p!}\right. \\ &\qquad\left.\times\ \ \big(\mathrm{vec}\big(\boldsymbol{X}\boldsymbol{W}_\varphi\big) - \mathrm{vec}\big(\boldsymbol{X}\boldsymbol{W}_\varphi\big)_0\big)^p\right) \end{aligned} \tag{6}$$

where $f_\theta^{(p)}$ denotes the pth-order partial derivative of $f(\cdot)$ with respect to $\mathrm{vec}(\boldsymbol{X}\boldsymbol{W}_\theta)$; $f_\varphi^{(p)}$ denotes the pth-order partial derivative of $f(\cdot)$ with respect to $\mathrm{vec}(\boldsymbol{X}_{NP}\boldsymbol{W}_\varphi)$. Since the variables of $f_\theta^{(p)}$ and $f_\varphi^{(p)}$ are the same, and the initial parameters of $\mathrm{vec}(\boldsymbol{X}\boldsymbol{W}_\theta)_0$ and $\mathrm{vec}(\boldsymbol{X}_{NP}\boldsymbol{W}_\varphi)_0$ are set to 0 in the network, the two items of 6 can be combined as

$$\begin{aligned} &f\big(\mathrm{vec}(\boldsymbol{X}\boldsymbol{W}_\theta), \mathrm{vec}\big(\boldsymbol{X}_{NP}\boldsymbol{W}_\varphi\big)\big) \\ &\quad\approx \sum_{p=0}^{P}\left(\frac{f^{(p)}\big(\mathrm{vec}(\boldsymbol{X}\boldsymbol{W}_\theta), \mathrm{vec}\big(\boldsymbol{X}_{NP}\boldsymbol{W}_\varphi\big)\big)}{p!}\right. \\ &\qquad\left.\times\ \ \big(\mathrm{vec}(\boldsymbol{X}\boldsymbol{W}_\theta)\mathrm{vec}\big(\boldsymbol{X}_{NP}\boldsymbol{W}_\varphi\big)\big)^p\right). \end{aligned} \tag{7}$$

When P is determined, $f\big(\mathrm{vec}(\boldsymbol{X}\boldsymbol{W}_\theta), \mathrm{vec}\big(\boldsymbol{X}_{NP}\boldsymbol{W}_\varphi\big)\big)$ has a closed-form solution. Let

$$\alpha_p = \frac{f^{(p)}\big(\mathrm{vec}(\boldsymbol{X}\boldsymbol{W}_\theta), \mathrm{vec}\big(\boldsymbol{X}_{NP}\boldsymbol{W}_\varphi\big)\big)}{p!} \tag{8}$$

which is the pth-order Taylor expansion coefficient of the nonlocal-aware pyramid feature maps. α_p can be computed once the function $f(\cdot)$ is known. Taking dot-product, the simplest form of $f(\cdot)$, for example, the output of the attention module can be expressed as

$$\begin{aligned} &\mathrm{vec}(\boldsymbol{Y}) \\ &\approx \boldsymbol{V}_\theta {\boldsymbol{V}_{\varphi NP}}^T \boldsymbol{V}_{gNP} \\ &\quad\times\begin{cases} \boldsymbol{V}_\theta = [\alpha_0\mathrm{vec}(\boldsymbol{X}\boldsymbol{W}_\theta)^0, \ldots, \alpha_P\mathrm{vec}(\boldsymbol{X}\boldsymbol{W}_\theta)^P] \\ \boldsymbol{V}_{\varphi NP} = [\alpha_0\mathrm{vec}(\boldsymbol{X}_{NP}\boldsymbol{W}_\varphi)^0, \ldots, \alpha_P\mathrm{vec}(\boldsymbol{X}_{NP}\boldsymbol{W}_\varphi)^P] \\ \boldsymbol{V}_{gNP} = \mathrm{vec}(\boldsymbol{X}_{NP}\boldsymbol{W}_g). \end{cases} \end{aligned} \tag{9}$$

The correlation operation of two $NC\times NC$ dimensional matrices is simplified to the correlation operation of an $NC\times(P+1)$ dimensional matrix and a $3NC\times(P+1)$ dimensional matrix.

According to the associative law of multiplication, calculating the last two items $\boldsymbol{V}_{\varphi gNP} = {\boldsymbol{V}_{\varphi NP}}^T\boldsymbol{V}_{gNP} \in \mathbb{R}^{P+1}$ first and then calculating $\boldsymbol{V}_\theta\boldsymbol{V}_{\varphi gNP} \in \mathbb{R}^{NC\times 1}$ can further reduce the computational complexity. This is because the complexity of ${\boldsymbol{V}_{\varphi NP}}^T\boldsymbol{V}_{gNP}$ is $O(3NC(P+1))(P \ll NC)$, which is less than $O((NC)^2)$, the complexity of $\boldsymbol{V}_{\theta\varphi}\boldsymbol{V}_g$ in the original nonlocal module. And the complexity of $\boldsymbol{V}_\theta\boldsymbol{V}_{\varphi gNP}$ is $O(NC(P+1))$, which is also less than, the complexity of $\boldsymbol{V}_\theta{\boldsymbol{V}_\varphi}^T$ in the original nonlocal module. Therefore, the complexity of our NP-Attention is $O(4NC(P+1))$, while the complexity of the original nonlocal attention module is $O(2(NC)^2)$. Since $P \ll NC$, $4(P+1) \ll 2NC$, the designed NP-Attention method is less complex and more efficient.

It should be noted that, since the dot-product function is used in this work, its Taylor expansion order $P = 1$, which is a fixed value. For other forms of $f(\cdot)$, the above formulas are also applicable, so a more general expression of P is adopted here. Regarding the setting of P, first, P is an intermediate variable, which is eliminated in the calculation of 9, and will not affect the establishment of the above equations. Second, the parameter P can usually be selected and adjusted according to the experiments and data sets. When using the same kernel function and controlling other conditions the same, the larger the P is, the more accurate and higher complexity the model is. The radial basis function (RBF) and Softmax forms of $f(\cdot)$ have been also validated. Experiments show that $P \le 3$ is more appropriate. When P is greater than 3, the accuracy improvement becomes very small and the complexity is higher. Due to the space limitations, only a more general proof is given here.

3) Edge Enhancement: The edge features of objects are also very important because the edge features of the same category are similar in top-view RS images. Therefore, learning from the idea of guided filtering, edge enhancement is considered in the Query of NP-Attention for guiding the Key-Value to focus more on edge features. The process of edge enhancement guided NP-Attention is illustrated in Fig. 3. This process can be expressed as

$$\begin{aligned} &\mathrm{vec}(\boldsymbol{Y}) \\ &\approx \boldsymbol{V}_{\theta E}{\boldsymbol{V}_{\varphi NP}}^T\boldsymbol{V}_{gNP} \\ &\quad\times\begin{cases} \boldsymbol{V}_{\theta E} = [\alpha_0\mathrm{vec}((\boldsymbol{E}+\boldsymbol{X})\boldsymbol{W}_\theta)^0, \ldots, \alpha_P\mathrm{vec}(\boldsymbol{X}\boldsymbol{W}_\theta)^P] \\ \boldsymbol{V}_{\varphi NP} = [\alpha_0\mathrm{vec}(\boldsymbol{X}_{NP}\boldsymbol{W}_\varphi)^0, \ldots, \alpha_P\mathrm{vec}(\boldsymbol{X}_{NP}\boldsymbol{W}_\varphi)^P] \\ \boldsymbol{V}_{gNP} = \mathrm{vec}(\boldsymbol{X}_{NP}\boldsymbol{W}_g) \end{cases} \end{aligned} \tag{10}$$

where $\boldsymbol{E}$ is the feature maps that fuse the multidirectional edge features of the object adaptively. $\boldsymbol{V}_{\theta E} \in \mathbb{R}^{NC\times(P+1)}$ denotes the Pth-order Taylor expansion of the edge-enhanced feature vectors (Query). After the introduction of $\boldsymbol{E}$, the complexity of $f(\cdot)$ is still $O(NC(P+1))$.

Because the edge feature extraction process should be implemented with convolutional layers and seamlessly embedded in the neural network, a simple operator, i.e., the Sobel

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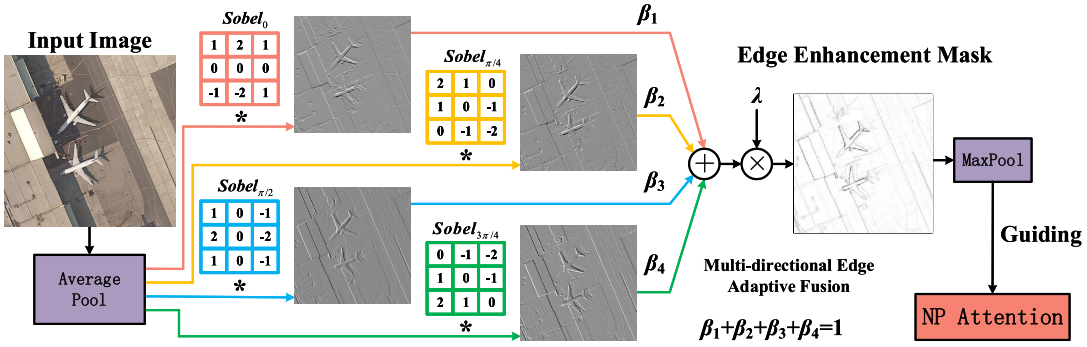

Fig. 3. Process of edge enhancement guided NP-Attention.

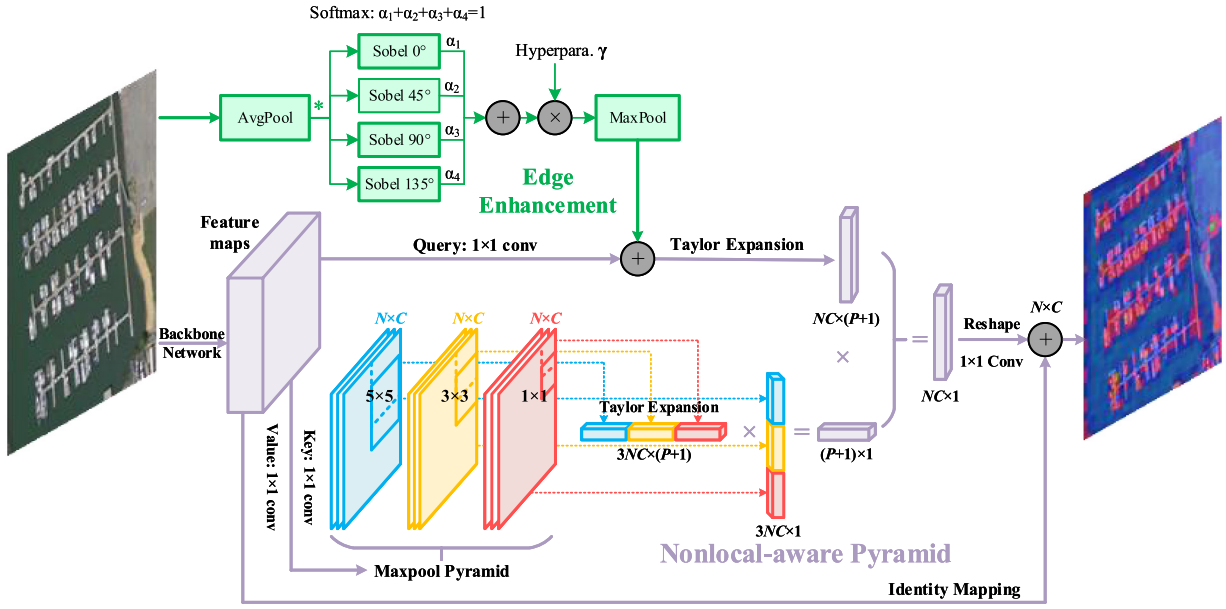

Fig. 4. Structure of the proposed NP-Attention in OD tasks.

operator, is selected to extract the edge features, as shown in Fig. 3. In RS images, the rotation angles of the objects are diverse, although the edge features of the same category objects are similar. Therefore, the Sobel operator is extended to four directions to extract the edge features with different rotation angles. Furthermore, the gray gradients of different rotation angles are fused adaptively during the model training process to obtain higher-quality edge features. The nonnegative parameters $\beta_1, \beta_2, \beta_3, \beta_4$ (where $\beta_1 + \beta_2 + \beta_3 + \beta_4 = 1$) are the weight coefficients of the Sobel convolution in different directions. Due to the fusion of Sobel features in four directions, the hyperparameter λ is set to 4.

4) Structure of NP-Attention: The structure and details of the proposed NP-Attention, consisting of a nonlocal-aware pyramid submodule and an edge enhancement submodule, are shown in Fig. 4. Similar to other Attention methods, an identity mapping of the input is added to NP-Attention for residual learning

$$\boldsymbol{Z} = \mathrm{GN}(\mathrm{shape}(\mathrm{vec}(\boldsymbol{Y}_E))\boldsymbol{W}_z) + \boldsymbol{X} = \mathrm{GN}(\boldsymbol{Y}_E\boldsymbol{W}_z) + \boldsymbol{X} \quad (11)$$

where $\boldsymbol{Z}$ is the output of NP-Attention; GN$(\cdot)$ denotes the group normalization [50], which divides the channels into groups for more efficient calculation in the neural network and computes the mean and variance for normalization within each group. The shape$(\cdot)$ operation indicates that the feature vector $\mathrm{vec}(\boldsymbol{Y}_E) \in \mathbb{R}^{NC\times 1}(N = rows \times cols)$ is reshaped to $\mathbb{R}^{rows\times cols\times C}$ (the original shape). The matrix $\boldsymbol{W}_z \in \mathbb{R}^{C\times C}$ is the weight matrix of the 1×1 convolution with nonlinear activation, which is used for integrating NP-Attention into the neural network and increasing the nonlinear characteristics, while keeping the size of the feature map unchanged.

B. Proposed Multiscale Refinement FPN

Large scale variations of objects may lead to misalignments between the scale of features extracted and the scale of real objects. In view of this, FPN [21] designed a top-down pyramid structure, as shown in Fig. 5(a). But the FPN does not propagate the high-resolution features from the shallower layers to the deeper layers, so that the large-scale branch of FPN,

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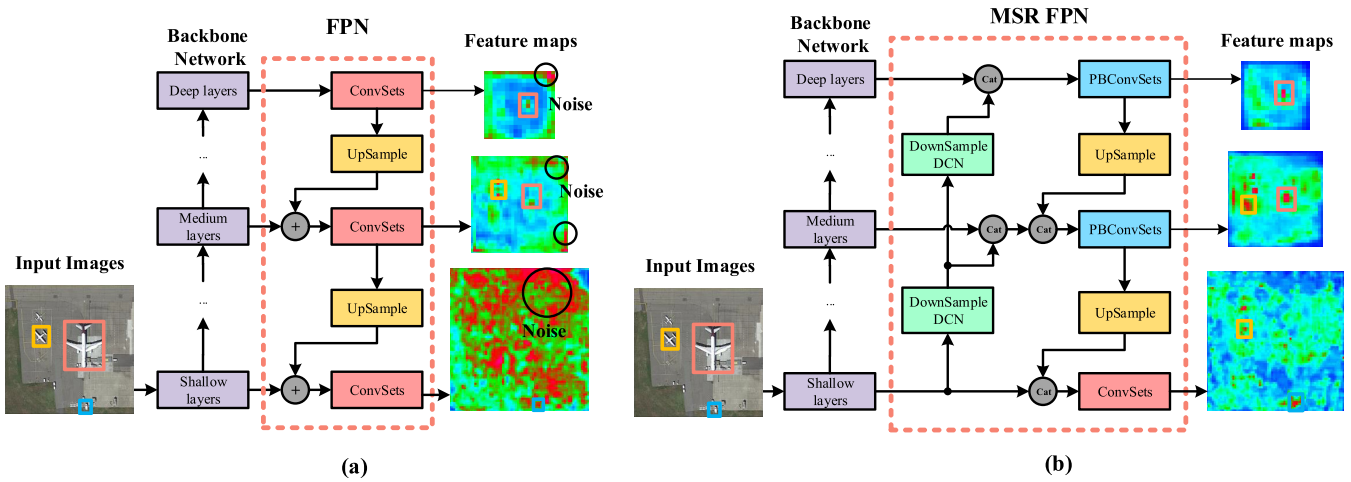

Fig. 5. Structure of proposed MSR-FPN in OD tasks. (a) Structure of FPN. (b) Structure of MSR-FPN.

represented by red circles in Fig. 5(a), cannot fuse features from other scale branches [38]–[40]. Besides, the range of CNN's receptive field is not large enough when detecting large objects, which is also one of the reasons that the detection accuracy of large objects is not noticeably improved. As shown in Fig. 5, there are many wrong features in the corners of the large-scale feature maps, so even if FPN improves the accuracy of small OD, it also causes a lot of false alarms.

In response to the above problems, MSR-FPN is proposed, and its structure is shown in Fig. 5(b). The difference between the FPN and MSR-FPN structures is mainly at two points: 1) new feature fusion pathway [the green blocks and their connected path in Fig. 5(b)] and 2) parallel branches with multiscale receptive fields [the blue blocks in Fig. 5(b)].

1) New Feature Fusion Pathway of MSR-FPN: As shown in Fig. 5(b), in order to propagate the high-resolution features from the shallower layers to the deeper layers, a bottom-up branch (green blocks and their connected path) is added, which allows each branch to fully fuse multiscale context features. It can be described as

$$\begin{aligned} \boldsymbol{F}_1 &= \text{Concat}(\boldsymbol{B}_1, \boldsymbol{UF}_2) \\ \boldsymbol{F}_{i\ (i=2,\ldots \boldsymbol{I}_{\max})} &= \text{Concat}(\boldsymbol{DF}_i, \boldsymbol{B}_i, \boldsymbol{UF}_i) * \boldsymbol{ConvSet} \\ \boldsymbol{F}_{\boldsymbol{I}_{\max}} &= \text{Concat}\left(\boldsymbol{DF}_{\boldsymbol{I}_{\max}-1}, \boldsymbol{B}_{\boldsymbol{I}_{\max}}\right) * \boldsymbol{ConvSet} \\ &\times \begin{cases} \boldsymbol{DF}_1 = \boldsymbol{B}_1 \\ \boldsymbol{DF}_{i\ (i=2,\ldots \boldsymbol{I}_{\max})} = \text{Down}(\boldsymbol{DF}_{i-1}) \\ \boldsymbol{UF}_{i\ (i=1,\ldots \boldsymbol{I}_{\max-1})} = \text{Up}(\boldsymbol{UF}_{i+1}) \\ \boldsymbol{UF}_{\boldsymbol{I}_{\max}} = \boldsymbol{B}_{\boldsymbol{I}_{\max}} \end{cases} \end{aligned} \tag{12}$$

where $\boldsymbol{F}_i$ denotes the feature map of the ith branch, $\boldsymbol{B}_i$ denotes the feature maps of the backbone connected with the feature pyramid, Up(·) is the up-sampling operation, and $\boldsymbol{ConvSet}$ is a series of convolutions. $\boldsymbol{UF}_i$ denotes the ith up-sampling operation, and $\boldsymbol{DF}_i$ denotes the ith down-sampling operation.

Unlike FPN, concatenation is adopted instead of adding lateral connections to retain as much multiscale information as possible. MSR-FPN enables the low-resolution features containing high-level semantic information and the high-resolution features containing low-level semantic information to propagate in the network so that each layer can fully integrate multiscale context semantics.

In order to compensate the position offsets generated in the down-sampling process, the down-sampling operation Down(·) adopts a deformable convolution with a 3×3 convolutional kernel and a Blurpool [51] with stride size of 2. The up-sampling operation Up(·) first employs 1×1 convolution to adjust the channel dimension, and then uses bilinear interpolation for up-sampling. $\boldsymbol{ConvSet}$ contains two sets of 1×1 and 3×3 convolutions.

The difference among the proposed MSR-FPN and PANet [38] and MFPN [40] is that in MSR-FPN an extra bottom-up pyramid is not added after the top-down pyramid, a calculation that is very expensive. The bottom-up structure is directly introduced in the top-down pyramid so that the $\boldsymbol{ConvSet}$ for multiscale feature fusion is shared by the two pathways.

2) Parallel Branches With Multiscale Receptive Fields: Whether the receptive field of each branch is matched with the object scale may also affect detection accuracy. Because the scale change of large objects is greater than that of small objects, the large-object branches of FPN are more severely affected by this problem.

In view of this, a series of parallel branches with multiscale receptive fields in the $\boldsymbol{ConvSet}$ is designed, as shown in Fig. 6. This structure uses dilated convolutions with different dilation rates to expand the multiscale receptive field of the convolutional layers. In accordance with the statistics of training samples' sizes, the dilation rates on the large-scale branch are set to 2, 4, and 6, and the dilation rates on the middle-scale branch are set to 2 and 4.

Multiple branches are the paralleled tree structure, and the number of feature maps in each branch is proportionally reduced (the sum of their numbers is the same as the original number of the feature maps in FPN) to improve efficiency. After introducing the dilation rate dila, the receptive field of the mth convolutional layer changes from

$$\text{RF}_m = \text{RF}_{m-1} + (\text{kernel}_m - 1) \times \prod_{i=1}^{m-1} \text{stride}_i \tag{13}$$

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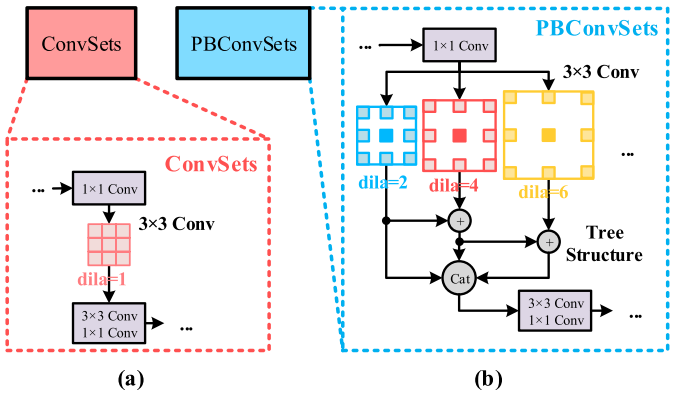

Fig. 6. Structure of proposed parallel branch ConvSets with multiscale receptive fields. (a) Structure of ConvSets. (b) Structure of PBConvSets.

to

$$\mathrm{RF}_m = \mathrm{RF}_{m-1} + (\mathrm{kernel}_m + (\mathrm{kernel}_m - 1) \times (\mathrm{dila} - 1) - 1) \times \prod_{i=1}^{m-1} \mathrm{stride}_i \quad (14)$$

where RF_m denotes the receptive field of the mth layer, kernel_m is the kernel size of the mth layer, and stride_i is the step length of the convolution. Moreover, drawing on the idea of attention, dynamic adjustment factors are designed into each parallel branch to form receptive field branch attention, and those branches with a suitable dilation rate are adaptively selected for feature fusion through the network training.

Benefiting from the above improvements, MSR-FPN fuses multiscale features from different layers of the network and adjusts suitable receptive fields with parallel branches consisting of multiscale dilated convolutions, which achieves the multiscale refinement.

C. Proposed MTR-Head

OD can be decomposed into localization and classification subtasks, and requires different features. For an one-stage detector, although performing two subtasks on the same convolutional layer has the advantages of simple structure and fast speed, these features of the different subtasks are more likely to interfere with each other, as shown in Fig. 7(a). They cause feature misalignment between the localization subtask, which focuses on the transformation equality of features, and the classification subtask, which focuses on the transformation invariance of features. Moreover, ordinary convolution can only extract the features of a fixed neighborhood with poor robustness of deformation.

Therefore, MTR-Head is developed based on the principle of "divide and conquer," as shown in Fig. 7(b). MTR-Head uses deformable convolution to predict the offsets of the convolution, introduces the modulation mechanism combined with an attention mask to highlight key features, and dynamically adjusts the feature extraction range of the convolution based on the shape of the object. Furthermore, the prediction of offsets and the feature extraction are jointly supervised by localization and classification subtasks. It can also be seen from the visualized features in Fig. 7(b) that when using MTR, there is less interference between the features of the two subtasks, and the extracted features own higher quality.

1) Network Structure of MTR-Head: The network structure of MTR-Head is shown in Fig. 8. The confidence prediction task is directly implemented with common convolutional layers like the original detection head (the output result uses sigmoid normalization). The object localization and classification tasks are implemented by two parallel deformable convolutions branches [52], [53], respectively. Based on the output features of MSR-FPN, the specific features that match the corresponding tasks are extracted. The modulation mechanism is used to provide another degree of freedom for the network module to adjust its spatial support area and improve the quality of the extracted features. Because of the correlation between the features of these two tasks, the offset module parameters of the two tasks are shared.

The output $\boldsymbol{H}_{\mathrm{out}}$ of each pixel on the feature map is

$$\boldsymbol{H}_{\mathrm{out}} = \sum_{k=1}^{K} w_k \cdot \boldsymbol{H}_{\mathrm{in}}(\mathrm{pos} + \mathrm{pos}_k + \Delta\mathrm{pos}_k) \cdot \Delta m_k \quad (15)$$

where $\boldsymbol{H}_{\mathrm{out}}$ and $\boldsymbol{H}_{\mathrm{in}}$ are the input and output features at the position pos, respectively. Suppose a convolution kernel with K sampling points (a 3×3 convolution kernel is used, so $K = 9$): w_k is the weight value of the kth sampling point; pos_k is the predefined offset of the kth sampling point; taking a 3×3 kernel as an example, $\mathrm{pos}_k \in \{(-1,-1),(-1,0),\ldots,(1,1)\}$. The term $\Delta\mathrm{pos}_k$ is the position offset of the learned kth sampling point. Because it is a decimal, Pin is obtained by bilinear interpolation. The term denotes the modulation factor of the kth sampling point (range between 0 and 1). The modulation mask is also enhanced by the mask of the NP-Attention. The proposed MTR-Head uses the prediction of localization and classification to directly supervise the learning of the offsets by the loss functions, which makes the learning of deformable convolution interpretable.

2) Loss Function: For supervised learning of multiscale object localization, a scale adaptive loss function with the adaptive scale factor $\xi_{\mathrm{scale}} = \chi - (\chi - 1) \cdot (g_w g_h / \mathrm{area})$, $(\chi \in \mathbb{Z}^+, \chi \geq 2)$ is improved based on the CIoU [49]

$$\mathrm{Loss}_{\mathrm{loc}} = \lambda_{\mathrm{loc}} \sum_{i=1}^{W} \sum_{j=1}^{H} 1^{\mathrm{obj}} \cdot \xi_{\mathrm{scale}} \cdot (1 - \mathrm{CIoU}) \quad (16)$$

where hyperparameter λ_{loc} is the weight of localization loss, and W and H represent the numbers of sampling points on the image width and height directions. The term area is the area of the input image. The function 1^{obj} is a symbolic function: when there is an object at the sampling point, $1^{\mathrm{obj}} = 1$, otherwise $1^{\mathrm{obj}} = 0$, and g_w and g_h are the width and height of the ground truth, respectively.

When the scale of the object relative to the input image is small, the scale factor ξ_{scale} approaches χ, and the localization loss is enlarged; when the scale of the object relative to the input image is large, the scale factor approaches 1, and the localization loss is not enlarged. In doing so, we fully consider the fact that small-scale objects are more sensitive to localization errors than large-scale objects. Therefore, the scale factor

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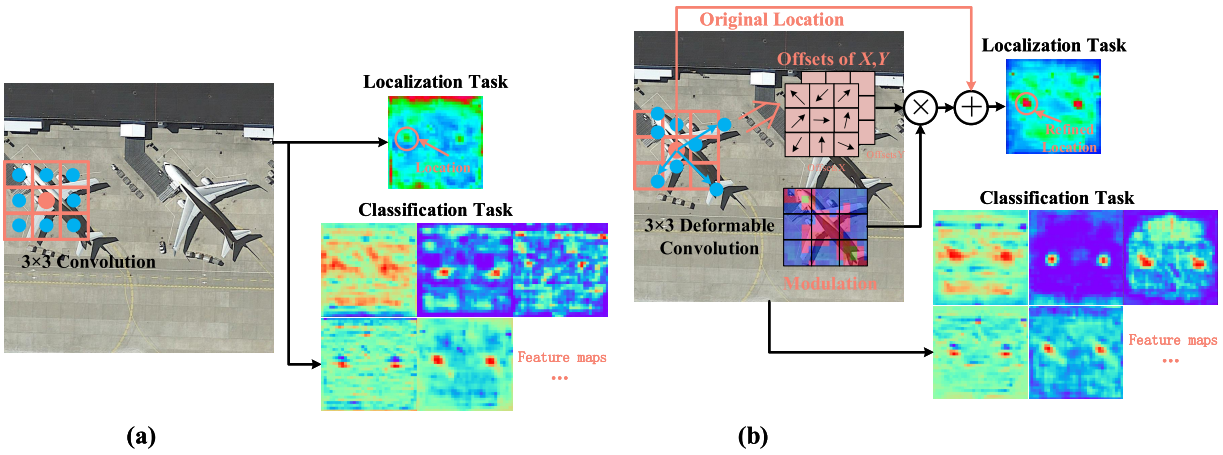

Fig. 7. Illustration of comparison between the ordinary detection head and the proposed MTR-Head. (a) Ordinary detection head. (b) MTR-Head.

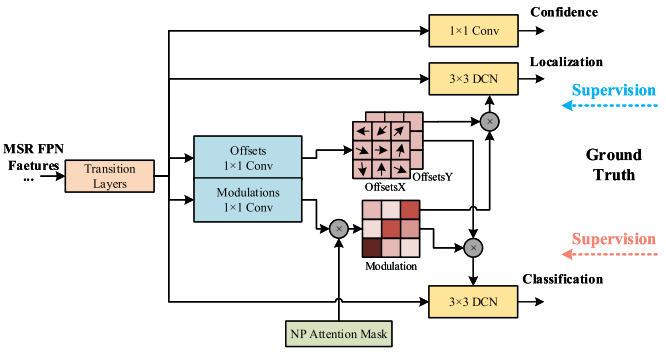

Fig. 8. Illustration of comparison between the ordinary detection head and the proposed MTR-Head.

ξ_{scale} allows the model to adaptively adjust the localization loss based on the object scale during the training process, which can improve the localization accuracy of multiscale objects. χ is set to 2. The larger the χ is, the larger the range of the scale factor is; that is, the greater the penalty of the object scale on the loss function is. When the scale of the objects in the image differs greatly, setting a larger χ value can usually get better results. The χ value is related to the instance scale of data sets, so a statistical analysis of the training samples is recommended to determine an appropriate χ value.

The confidence loss is presented as

$$\begin{aligned}\text{Loss}_{\text{conf}} = \lambda_{\text{obj}} \sum_{i=1}^{W}\sum_{j=1}^{H} 1^{\text{obj}}(-\vartheta(1-\sigma(\text{conf}_{i,j}))^{\gamma} \\ \times \log(\sigma(\text{conf}_{i,j})) + \lambda_{\text{noobj}} \sum_{i=1}^{W}\sum_{j=1}^{H} 1^{noobj} \\ \times(-\vartheta(1-\sigma(1-\text{conf}_{i,j}))^{\gamma} \\ \times\log(\sigma(1-\text{conf}_{i,j}))) \end{aligned} \tag{17}$$

where λ_{obj} and λ_{noobj} are the weights of the confidence loss with and without objects, respectively, and $\lambda_{\text{obj}}/\lambda_{\text{noobj}} = 5$. The values of hyperparameters ϑ and γ are set the same as those in the article [10]. Here, $\sigma(\cdot)$ is the sigmoid function, and $\text{conf}_{i,j}$ denotes the probability that there is an object at this sampling point.

The loss of the classification task is in the form of binary cross-entropy

$$\begin{aligned}\text{Loss}_{\text{cls}} = \lambda_{\text{cls}} \sum_{i=1}^{W}\sum_{j=1}^{H} 1^{\text{obj}} \cdot \left(-\sum_{l=1}^{C} p_{ij}^{\text{cls}_l} \cdot \log\left(\hat{p}_{ij}^{\text{cls}_l}\right) \right. \\ \left. +\left(1-p_{ij}^{\text{cls}_l}\right)\cdot\log\left(1-\hat{p}_{ij}^{\text{cls}_l}\right) \right) \end{aligned} \tag{18}$$

where λ_{cls} is the weight of confidence loss, C is the number of categories, $p_{ij}^{\text{cls}_l}$ denotes the probability that the ground truth object belongs to the category l, and $\hat{p}_{ij}^{\text{cls}_l}$ denotes the probability that the object is predicted to be category l. Therefore, the total loss function of the OD task is

$$\text{Loss} = \text{loss}_{\text{conf}} + \text{Loss}_{\text{loc}} + \text{Loss}_{\text{cls}}. \tag{19}$$

The final prediction model can be obtained by minimizing the training loss.

IV. Experiments and Discussions

In this section, experiments on two public RS data sets are conducted to verify the effectiveness of the proposed NPMMR-Det and further explore its detection performance. First, experimental conditions are described, including data sets, evaluation metrics, implementation details, and baseline. Second, each component of NPMMR-Det, including NP-Attention, MSR-FPN, and MTR-Head, is evaluated, compared with the benchmarking methods. Third, detailed ablation experiments are performed, and the impact of each improvement on the final detection performance is discussed. Finally, experiments comparing the proposed approach with state-of-the-art methods are conducted on public data sets, and the results are analyzed and visualized.

All experiments are implemented on a computer with an Intel Xeon E5-2678 v3 CPU (2.50 GHz), 64 GB of memory, and an NVIDIA GeForce RTX2080Ti GPU (11 GB memory) for acceleration.

A. Experimental Conditions

1) Data Sets: The DIOR data set and the DOTA data set, both widely used in the field of RSOD, are used for

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experiments and evaluation. The details of these data sets are as follows.

a) DIOR data set: DIOR [54] is a large aerial image data set for OD with 23 463 images containing a total of 190 288 instances. The size of all images is 800 × 800 pixels. Of these, 5862 images are used for training, 5863 images are used for validation, and 11 738 images are used for testing. These images cover 20 common object categories, including Airplane (c1), Airport (c2), Baseball field (c3), Basketball court (c4), Bridge (c5), Chimney (c6), Dam (c7), Expressway service area (c8), Expressway toll station (c9), Golf course (c10), Ground track field (c11), Harbor (c12), Overpass (c13), Ship (c14), Stadium (c15), Storage tank (c16), Tennis court (c17), Train station (c18), Vehicle (c19), and Windmill (c20).

b) DOTA data set: DOTA data set [55] is another large aerial image data set for OD. It contains a total of 2806 aerial images, whose size range from 800 × 800 pixels to 4000 × 4000 pixels. Of these, 1418 images are used for training, 453 images are used for validation, and 947 images are used for testing. The DOTA data set covers 15 object categories, which are defined as Plane (PL), Baseball diamond (BD), Bridge (BR), Ground field track (GFT), Small vehicle (SV), Large vehicle (SV), Ship (SH), Tennis court (TC), Basketball court (BC), Storage tank (ST), Soccer-ball field (SBF), Roundabout (RA), Harbor (HA), Swimming pool (SP), and Helicopter (HC).

Due to the huge and different sizes of the images in the DOTA data set, it is difficult to directly input them into the neural network models for training. It should be noted that since the DOTA data set does not give a uniform standard for image cropping, different articles crop the image into subimages of different sizes. This difference in training data eventually leads to a certain difference in OD performance. To alleviate this effect as much as possible, we present some parameters below. We crop the large images into 608 × 608 pixel subimages with an overlap of 100 pixels (this is because an object may be split when cropping. To ensure that an object can be fully presented in a subimage, the overlap is used). The threshold determines whether to keep the instance if the instance is cut down in the process of split is 0.7.

2) Evaluation Metrics: The mAP (mean Average Precision) is used to quantitatively evaluate detection accuracy; mAP has been widely used in many detection tasks. The mAP is obtained by averaging the AP (Average Precision) of each object category. The AP is the area under the P-R curve (Precision-Recall curve) of each category. The precision and recall can be, respectively, formulated as follows:

$$\text{precision} = \text{TP}/(\text{TP} + \text{FP}) \tag{20}$$

$$\text{recall} = \text{TP}/(\text{TP} + \text{FN}) \tag{21}$$

where TP, FP, and FN denote the numbers of true positives, false positives, and false negatives, respectively. According to the PASCAL VOC metrics widely used in the field of OD, if the Intersection over Union (IoU) between the predicted bounding box and the ground truth is greater than 0.5, the instance predicted is considered to be TP. It should be noted that the mAP is usually expressed as a percentage, and the convention in the field of OD is to omit the percentage symbol in the results. The detection speed is evaluated by the average time detecting an image and the frames per second (fps). The model size is evaluated by the space occupied by the weight parameters.

3) Implementation Details: To compare the proposed NPMMR-Det with state-of-the-art methods more fairly, training hyperparameters are set to be the same or similar to those of the state-of-the-art methods compared. The initial learning rate is 1×10^{-4}, the final learning rate is 1×10^{-6}, and the learning rate is updated using a cosine update strategy. The weight decay is 5×10^{-4}, and momentum is 0.9. The maximum training epoch is 180. The IoU threshold is 0.5, the confidence threshold is 0.05, and the NMS threshold is 0.5. The initial weight of the backbone is pretrained on the ImageNet data set.

4) Selection of the Baseline Model: Since the proposed method is a one-stage detector, the YOLOv3 [56] is selected as the baseline of the ablation experiments. Moreover, the standard test results of YOLOv3 have been reported using DIOR data set [54]. The mAP of YOLOv3 on the DIOR data set published in [54] is 57.1. Considering the difference between the experimental equipment used in the baseline and our work, which may cause additional errors in the results, YOLOv3 is reimplemented, and many improvements proposed in the latest research [57] are added to the baseline, resulting in an increase in the mAP of a baseline of 68.69.

B. Evaluations of Each Component of NPMMR-Dets

The NP-Attention, MSR-FPN, and MTR-Head are evaluated and compared with the benchmarking methods to verify their effectiveness and advancement.

1) Evaluations of the Proposed NP-Attention: The NP-Attention Module consists of the nonlocal-aware pyramid submodule and the edge enhancement submodule. Their performance will be evaluated separately. The state-of-the-art attention methods, SENet [25], Nonlocal Network [20], GCNet [27], and CGNL [29], are selected as the control groups. Among them, SENet [25] is the channel attention method, Nonlocal Network [20] is the spatial attention method, GCNet [27] and CGNL [29] are space-channel hybrid attention methods.

The evaluation results of the nonlocal-aware pyramid submodule are listed in Table I. It is clearly to observe that the mAP and the $\Delta\text{mAP}/\Delta\text{Time}$ of the nonlocal-aware pyramid (NP) are the best. The NP submodule increases the model parameters only slightly. This indicates that the NP submodule not only improves mAP the most compared to benchmarking methods, but also has the highest conversion efficiency from the time to the mAP. It is generally known that there is usually a trade-off between detection time and mAP, while the NP submodule has the highest mAP increase per unit time increase. In addition, from the experimental results of different kernel functions with different values of P, when using the same kernel function and controlling other conditions the same, the larger the P is, the more accurate and

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TABLE I

EVALUATION OF THE NONLOCAL-AWARE PYRAMID SUBMODULE

Methods	mAP	Inference Time (s/img)	$\Delta mAP/\Delta Time$	Model Size (MB)
Baseline	68.69	**0.02301**	-	**235**
Nonlocal Network [20]	69.69 (+1.00)	0.02931 (+0.00630)	158.73	250
SENet [25]	68.93 (+0.24)	0.02336 (+0.00035)	685.71	236
GCNet [27]	69.81 (+1.12)	0.02533 (+0.00232)	482.76	238
CGNL [29]	69.78 (+1.09)	0.02487 (+0.00186)	586.02	236
NP (Softmax, P=3)	70.06 (+1.37)	0.02591 (+0.00290)	472.41	236
NP (RBF, P=3)	70.12 (+1.43)	0.02536 (+0.00235)	608.51	236
NP (RBF, P=1)	70.03 (+1.34)	0.02522 (+0.00221)	606.33	236
NP (Dot-product, P=1)	**70.20 (+1.51)**	0.02501 (+0.00200)	**755.00**	236

Note: Bold font indicates the best results.

TABLE II

EVALUATION OF THE NONLOCAL-AWARE PYRAMID SUBMODULE

Methods	mAP	Inference Time (s/img)	$\Delta mAP/\Delta Time$	Model Size (MB)
Baseline	68.69	**0.02301**	-	**235**
Nonlocal Network [20]	69.69 (+1.00)	0.02931 (+0.00630)	158.73	250
Nonlocal Network [20] + EE	69.69 (+1.21)	0.02935(+0.00634)	190.85	250
CGNL [29]	69.78 (+1.09)	0.02487 (+0.00186)	586.02	236
CGNL [29] + EE	70.13 (+1.44)	0.02488 (+0.00187)	770.05	236
NP	70.20 (+1.51)	0.02501 (+0.00200)	755.00	236
NP + EE	**70.37 (+1.68)**	0.02509 (+0.00208)	**807.69**	236

Note: Bold font indicates the best results.

higher complexity the model is. As for the higher accuracy of using the Dot-product (linear) function, it may be because the Taylor expansion of the linear function is an accurate solution rather than an approximate solution like using the RBF function. What needs to be explained is that it does not mean that the mAP of using the Dot-product function is always better than that of using other functions in any case, it is the results of our current exploration of the specific data set.

The evaluation of the edge enhancement (EE) submodule is shown in Table II. For evaluating the performance of introducing the EE submodule fairly, it is not only introduced in NP-Attention but also introduced in several commonly used attention methods, which act as the control groups. It can be seen from Table II that the mAP and the ΔmAP/ΔTime of the nonlocal-aware pyramid with edge enhancement (NP+EE) are the highest. This demonstrates the effectiveness of introducing the EE submodule into the NP-Attention. Moreover, the EE submodule hardly brings about any increase in model parameters. These demonstrate the effectiveness and advancement of NP-Attention, which consists of the NP submodule and the EE submodule.

NP-Attention is a plug-and-play module that can be used in the backbone network. To compare performance when embedding it in different positions of the CNN, the experiments shown in Table III are conducted. The backbone network used by NPMMR-Det is Darknet53 [56], which consists of five residual blocks. Since the last three residual modules are connected to our MSR-FPN to extract the multiscale features, which directly affects the final predictions, the performance of embedding NP-Attention into these three blocks is evaluated. The results show that embedding NP-Attention into the ResBlock4, which connects with the medium branch of MSR-FPN, yields the highest mAP and the best balance between the mAP and detection time. This may be because embedding NP-Attention at this position can make the multiscale nonlocal dependencies most efficiently used by each branch of MSR-FPN.

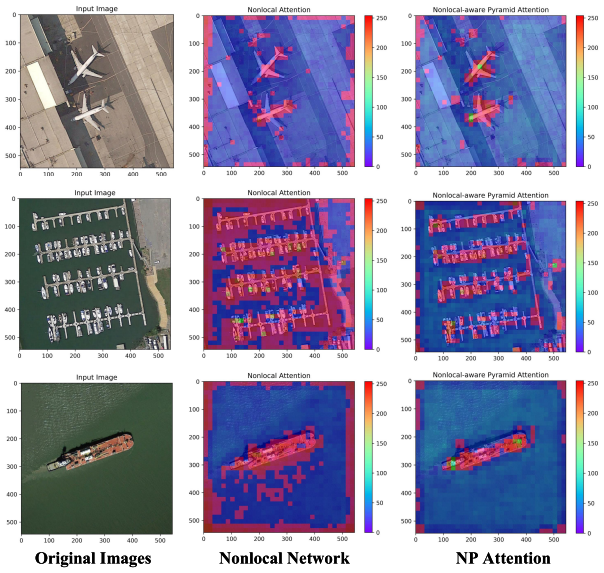

Fig. 9. Heat Maps of the Nonlocal Network and NP-Attention.

In addition, the visual attention heat maps of the Nonlocal Network [20] and the NP-Attention proposed are shown in Fig. 9. The subimages in the first column are the original input images, the subimages in the second column are the heat maps of Nonlocal Attention [20], and the subimages in the third column are the heat map of our NP-Attention. The closer to the red areas in the heat maps are, the higher the attention weight.

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TABLE III
EVALUATION OF EMBEDDING NP-ATTENTION INTO DIFFERENT POSITIONS OF THE NETWORK

Methods	mAP	Inference Time (s/img)	$\Delta mAP/\Delta Time$	Model Size (MB)
Baseline	68.69	**0.02301**	-	**235**
in the ResBlock3	69.71 (+1.02)	0.02462 (+0.00131)	778.63	236
in the ResBlock4	**70.37 (+1.68)**	0.02509 (+0.00208)	**807.69**	236
in the ResBlock5	70.06 (+1.37)	0.02584 (+0.00283)	484.10	240

Note: Bold font indicates the best results.

TABLE IV
EVALUATION OF THE PROPOSED MSR-FPN WITH COMPARISON WITH OTHER DETECTORS

Methods	mAP	Inference Time (s/img)	$\Delta mAP/\Delta Time$	Model Size (MB)
Baseline (FPN [21])	68.69	0.02301	-	235
PANet [38]	70.07 (+1.38)	0.02788 (+0.00487)	283.37	268
MFPN [40]	70.09 (+1.40)	0.02829 (+0.00528)	265.15	273
MSR-FPN	70.15 (+1.44)	0.02519 (+0.00218)	**660.55**	264
MSR-FPN ($\boldsymbol{Convset}$×2)	**70.44 (+1.85)**	0.02702 (+0.00401)	461.35	280
MSR-FPN (Separable $\boldsymbol{Convset}$×2)	69.05 (+0.36)	**0.02287 (-0.00014)**	-	**231**

Note: Bold font indicates the best results.

Compared with the object areas focused on the heat maps of the Nonlocal Network (the second column), the object areas (especially the center positions of the objects) are focused more accurately and the irrelevant areas are suppressed more effectively in the heat maps of NP-Attention (the third column). This suggests that the network introducing NP-Attention can extract higher-quality features than the original network and the Nonlocal Network. It demonstrates the effectiveness of NP-Attention improvements for solving the misalignment problem between the network's emphasis and the objects from another aspect.

2) Evaluations of the Proposed MSR-FPN: FPN [21] and PANet [38] and MFPN [40], the latest improvements on FPN, are used as the control group. Among them, FPN [21] is the baseline, PANet [38] and MFPN [40] improve new network connection pathways based on it.

The evaluation results of the proposed MSR-FPN are provided in Table IV. The MSR-FPN further improves detection accuracy, and the increase in time complexity is slight (only a few milliseconds, or even less than this value of many commonly used methods). The efficiency conversion ratio, ΔmAP/ΔTime, is higher than that of the benchmarking methods.

For a high mAP, the number of convolutional layers in the ***ConvSets*** is doubled (the number of convolutional layers is as many as that of the PANet's double-pyramid structure). And the performance of MSR-FPN (***ConvSets*** $\times 2$) is better in both mAP and detection time. These results suggest that although the double-pyramid structure can improve accuracy, the doubled increase in model size with many redundant features will also increase computational cost and reduce detection efficiency. On the other hand, the value of shallow features is to contain more high-resolution details from the original image, and the structure of PANet [38], which introduces the bottom-up structure after the top-down structure, propagates the pure shallow features to the deeper layers that have been submerged in other, fused-together features. MSR-FPN avoids these problems and obtains better performance.

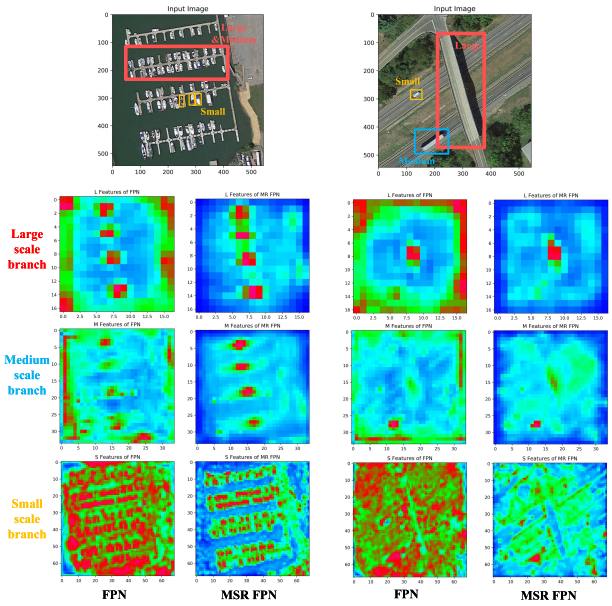

Fig. 10. Visualized Feature Maps of FPN and MSR-FPN.

In order to achieve faster detection speed, the convolutions of ***ConvSets*** are modified to separable convolutions. In this case, the detection accuracy of MSR-FPN is slightly higher than that of the baseline, but it takes less time than the baseline (because the ΔmAP/ΔTime is negative in this case, we do not calculate its value). The ΔmAP/ΔTime is used to measure the time cost of improving accuracy, and three schemes (the accuracy first, the time first, and the balance between the two) based on the same idea are given in the comparative experiments. These three MSR-FPN schemes can be selected according to different situations and applications. In the following experiments, the balance scheme is used as the control group.

The feature maps of MSR-FPN are visualized in Fig. 10. The subimages in the first row are the original input images.

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TABLE V
EVALUATION OF THE PROPOSED MST HEAD WITH MAP AND INFERENCE TIME

Methods	mAP	Inference Time (s/img)	Model Size (MB)
Baseline	68.69	**0.02301**	235
Baseline using the MTS-Head	70.09 (+1.40)	0.02405 (+0.00104)	235
Baseline using the MTS-Head + New loss	**70.37 (+1.68)**	0.02405 (+0.00104)	235

Note: The loss function is only used when training; it does not increase the detection time.

The objects with different scales are marked with rectangles of different colors. The subimages in the second, third, and fourth rows are output feature maps from different scale branches of the feature pyramids. The subimages in the first and third columns are feature maps of FPN, and the subimages in the second and fourth columns are feature maps of MSR-FPN. It can be observed from Fig. 10 that the effective features are found more accurately obtained by MSR-FPN. For example, the large-scale branch of MSR-FPN extracts the features of the large-scale objects (harbors and bridge) in the images, while the large-scale branch of FPN extracts the wrong features in the corner area, although it can also extract the objects' features. In the medium-scale branch, some features that should have been extracted by FPN are missing. These demonstrate the effectiveness of introducing the bottom-up structure to fuse multiscale features in the large and medium-scale branches in MSR-FPN. As another example, due to the multiscale receptive field refinement strategy, the feature maps of MSR-FPN have less noise than the feature maps of the FPN (especially the feature map of the small-scale branch in the fourth row). The multiscale feature extraction tasks can be more effectively distributed to each branch, and each branch extracts the object features with the corresponding scale. These experiments demonstrate the effectiveness of MSR-FPN from the perspective of feature extraction.

3) Evaluation of the Proposed MTR-Head: The evaluation results of the proposed MSR-FPN are listed in Table V. Using only MSR Head increases mAP by 1.40 while using MST Head with the new loss function improves the mAP by 1.68. Replacing the original detection head by MSR Head only slightly increases the detection time and does not increase additional model parameters. The new loss corresponding to the MSR Head is only used in the training phase and brings no additional inference time. These results indicate that MST Head can provide a significant mAP improvement with the little time cost.

The feature maps of an ordinary detection head and MTR-Head are visualized in Fig. 11. The subimages in the first column are the original input images; the subimages in the second column are the localization features; the subimages from the third, fourth, and fifth columns are the classification features. The subimages in different rows are the features from different scales. Fig. 11(a) shows that there are many false features (noise) in the feature maps of different scales when ordinary detection heads are used. This is caused by feature misalignment between the localization subtask, which focuses on the transformation equality of features, and the classification subtask, which focuses on the transformation invariance of features. It can be seen that, especially in the feature maps of a small scale, the mutual interference of different features is particularly serious. In contrast, Fig. 11(b) shows that this misalignment is refined and the interference between features of different tasks is greatly alleviated when the proposed MTR-Heads are used. This experiment demonstrates that the quality of feature maps is significantly improved by the proposed MTR-Head.

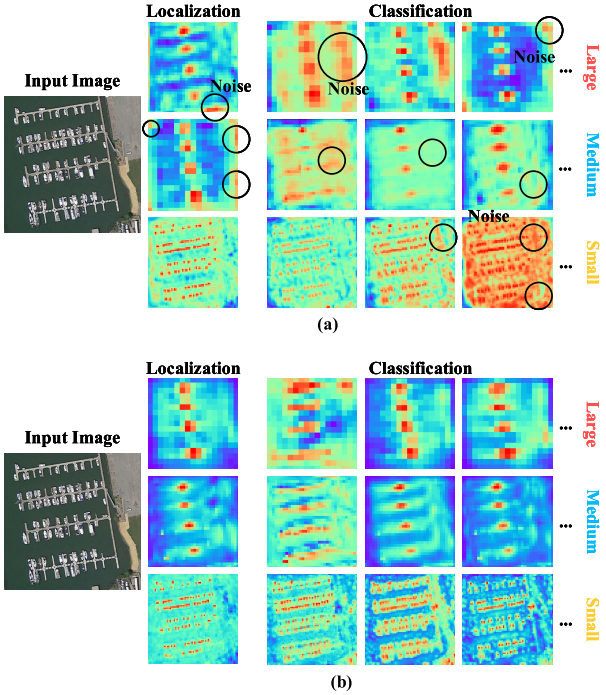

Fig. 11. Visualized Feature Maps, using an ordinary detection head and MTR-Head. (a) Features of ordinary detection head. (b) Features of MTR- Head.

C. Ablation Experiments and Discussions of NPMMR-Det

The ablation experiments are conducted on the DIOR data set [54] (currently the largest RSOD data set with the most object categories) to verify the effectiveness of the improvements proposed in this article and to test their performance. First, the ablation experiments of the entire framework are conducted to evaluate the impact of three improvements (i.e., NP-Attention, MSR-FPN, and MTR-Head) of the proposed NPMMR-Det. Then, like many current articles in the field of RSOD (for example, [17], [22], [58]), experiments using the training and validation samples and introducing the data augmentation, and adopting the multiscale testing are also conducted to get higher detection accuracy.

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TABLE VI
ABLATION EXPERIMENTS OF SELECTED MODULES FOR THE PROPOSED NPMMR-DET

Modules	Baseline	NP-Attention	MSR-FPN	MTR-Head	mAP	Inference Time (s/img)	Model Size (MB)
Selected Module(s)	✓				68.69	0.02301	235
	✓	✓			70.37 (+1.68)	0.02509	236
	✓		✓		70.15 (+1.46)	0.02519	264
	✓			✓	70.37 (+1.68)	0.02405	235
	✓	✓	✓		70.69 (+2.00)	0.02812	265
		✓	✓	✓	71.57 (+2.88)	0.02942	265

TABLE VII
MORE DETAILED RESULTS OF ABLATION EXPERIMENTS

Methods	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15	C16	C17	C18	C19	C20	mAP
Baseline	82.9	78.0	69.5	91.7	41.8	78.7	61.0	60.9	54.4	77.7	72.0	62.1	56.7	**89.8**	43.8	74.0	88.7	58.4	55.7	76.2	68.7
NP-Attention	83.0	79.9	69.8	91.1	42.3	79.6	61.1	**64.0**	56.3	78.7	75.2	60.8	56.0	**89.8**	67.3	73.3	88.6	56.8	56.1	77.6	70.4
MSR-FPN	83.3	79.4	69.6	91.1	41.7	78.3	63.7	63.4	56.3	77.9	75.3	61.7	56.0	89.2	61.1	74.1	87.7	60.5	54.9	77.8	70.1
MTR-Head	83.9	79.5	72.0	91.3	43.0	79.4	60.5	61.8	56.8	77.5	76.8	60.8	57.1	**89.8**	61.4	77.0	89.8	56.2	54.8	78.0	70.4
NP-Attention + MSR-FPN	80.8	81.9	71.2	91.2	41.5	78.8	**64.4**	**64.0**	56.7	80.1	77.2	61.1	57.2	89.5	**63.3**	74.5	88.6	58.0	54.7	**79.2**	70.7
NPMMR-Det	**86.4**	**82.2**	**71.6**	**91.8**	**44.6**	**79.3**	63.5	63.8	**56.8**	**80.8**	**77.6**	**62.5**	**58.0**	89.6	60.6	**77.3**	**88.9**	**61.8**	**56.6**	77.7	**71.6**

Note: Bold font indicates the best results. The object categories are explained as follows.

C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15	C16	C17	C18	C19	C20
Airplane	Airport	Baseball field	Basketball court	Bridge	Chimney	Dam	Expressway service area	Expressway toll station	Golf course	Ground track field	Harbor	Overpass	Ship	Stadium	Storage tank	Tennis court	Train station	Vehicle	Windmill

TABLE VIII
FURTHER ABLATION EXPERIMENTS USING THE PROPOSED NPMMR-DET FRAMEWORK

Methods	Training Data	Multi-scale Testing	mAP	Inference Time (s/img)
NPMMR-Det	Train	No	71.57	0.02942
NPMMR-Det	Train + Val	No	77.38 (+NP-Attention 74.90, +NP-Attention& MSR-FPN 76.62)	0.02942
NPMMR-Det*	Train	Yes	73.30	0.08680
NPMMR-Det*	Train + Val	Yes	78.13	0.08681

1) Evaluation of the Proposed NPMMR-Det Framework: The results of these ablation experiments are listed in Table VI. We find that NP-Attention improves mAP by 1.68 to 70.37, while the detection time and model size only increase slightly. The designed MSR-FPN improves mAP with 1.46 while the detection speed does not slow down significantly. Replacing the original detection head with the MTR-Head increases mAP by 1.68, without increasing the model parameters. And the increase of detection time caused by using MTR-Head is the least of the three improvements. When all the three parts are replaced, a novel detector (NPMMR-Det) is obtained with an mAP of 71.57. The above experimental results demonstrate that the three improvements proposed in this article can effectively increase the accuracy of detection without significantly increasing the detection time.

Comparing the last two rows of Table VI, the mAP increment of using MTR-Head is greater than that of using the other two. This may be due to the similarity effects of employing NP-Attention and MSR-FPN. The mAP of the proposed NPMMR-Det, comprising NP-Attention, MSR-FPN, and MTR-Head, reaches a maximum of 71.57, which is 2.88 higher than that of the new baseline, with a speed of 34.0 fps.

More detailed results of the ablation experiments are given in Table VII, which shows that the APs of different categories on the DIOR data set. It should be noted that the APs given by DIOR's benchmark are all reserved to one decimal place. For comparison, experimental results are also reserved to one decimal place. In Table VII, the optimal results are shown in bold. In these results, the proposed NPMMR-Det yields the best performance in the ablation experiments in 15 of the 20 categories, which further demonstrates NPMMR-Det's effectiveness and generality.

2) Further Ablation Experiments and Discussions: Like most other work in the OD field, for higher detection performance, both the training samples and the validation samples for model training (for the DIOR data set, the number of training samples has increased by one time) employ further data augmentations (including the Mixup and Label Smooth). The hyperparameters are no longer adjusted. Moreover, multiscale testing is also adopted to pursue ultimate accuracy. The results of further ablation experiments are listed in Table VIII.

After using more training samples and further data augmentations, the mAP of NPMMR-Det reaches 77.38. Specifically, the mAP is 74.90 when only the NP-Attention is used, and the mAP is 76.62 when both the NP-Attention and the MSR-FPN are used. Based on this, multiscale testing can make mAP reach the extreme of 78.13. This result is represented by NPMMR-Det*. However, multiscale testing seriously affects detection because the image pyramids are used. In this

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TABLE IX
COMPARATIVE EXPERIMENTS ON THE DIOR DATA SET

Methods	Year	Backbone	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15	C16	C17	C18	C19	C20	mAP	Speed (fps)
Faster R-CNN [5]	2015	VGG16	53.6	53.2	78.8	66.2	28.0	70.9	62.3	69.0	55.2	68.0	56.9	50.2	50.1	27.7	73.0	39.8	75.2	38.6	23.6	45.4	54.1	16.9
Faster R-CNN with FPN [21]	2017	ResNet50	54.1	71.4	63.3	81.0	42.6	72.5	57.5	68.7	62.1	73.1	76.5	42.8	56.0	71.8	57.0	53.5	81.2	53.0	43.1	80.9	63.1	15.5
		ResNet101	54.0	74.5	63.3	80.7	44.8	72.5	60.0	75.6	62.3	76.0	76.8	46.4	57.2	71.8	68.3	53.8	81.1	59.5	43.1	81.2	65.1	14.6
Mask R-CNN [60]	2017	ResNet50	53.8	72.3	63.2	81.0	38.7	72.6	55.9	71.6	67.0	73.0	75.8	44.2	56.5	71.9	58.6	53.6	81.1	54.0	43.1	81.1	63.5	14.0
		ResNet101	53.9	76.6	63.2	80.9	40.2	72.5	60.4	76.3	62.5	76.0	75.9	46.5	57.4	71.8	68.3	53.7	81.0	62.3	43.0	81.0	65.2	11.2
SSD [9]	2016	VGG16	59.5	72.7	72.4	75.7	29.7	65.8	56.6	63.5	53.1	65.3	68.6	49.4	48.1	59.2	61.0	46.6	76.3	55.1	27.4	65.7	58.6	**48.9**
YOLOv3 [56]	2018	Darknet53	72.2	29.2	74.0	78.6	31.2	69.7	26.9	48.6	54.4	31.1	61.1	44.9	49.7	87.4	70.6	68.7	87.3	29.4	48.3	78.7	57.1	44.5
RetinaNet [10]	2017	ResNet50	53.7	77.3	69.0	81.3	44.1	72.3	62.5	76.2	66.0	77.7	74.2	50.7	59.6	71.2	69.3	44.8	81.3	54.2	44.4	83.4	65.7	14.9
		ResNet101	53.3	77.0	69.3	85.0	44.1	73.2	62.4	78.6	62.8	78.6	76.6	49.9	59.6	71.1	68.4	45.8	81.3	55.2	45.1	85.5	66.1	13.6
PANet [38]	2019	ResNet50	61.9	70.4	71.0	80.4	38.9	72.5	56.6	68.4	60.0	69.0	74.6	41.6	55.8	71.7	72.9	62.3	81.2	54.6	48.2	86.7	63.8	8.4
		ResNet101	60.2	72.0	70.6	80.5	43.6	72.3	61.4	72.1	66.7	72.0	73.4	45.3	56.9	71.7	70.4	62.0	80.9	57.0	47.2	84.5	66.1	6.9
CornerNet [61]	2018	Hourglass104	58.8	84.2	72.0	80.8	46.4	75.3	64.3	81.6	76.3	79.5	79.5	26.1	60.6	37.6	70.7	45.2	84.0	57.1	43.0	75.9	64.9	7.0
CSFF [59]	2020	ResNet101	57.2	79.6	70.1	87.4	46.1	76.6	62.7	82.6	73.2	78.2	81.6	50.7	59.5	73.3	63.4	58.5	85.9	61.9	42.9	86.9	68.0	13.1
SCRDet++ (RetinaNet-based) [58]	2020	ResNet50	64.3	79.0	73.2	85.7	45.8	76.0	68.4	79.3	68.9	77.7	77.9	56.7	62.2	70.4	67.7	60.4	80.9	63.7	44.4	84.6	69.4	9.6
		ResNet101	71.9	85.0	79.5	88.9	52.3	79.1	**77.6**	89.5	77.8	84.2	83.1	64.2	65.6	71.3	76.5	64.5	88.0	70.9	47.1	85.1	75.1	8.1
SCRDet++ (FPN-based) [58]	2020	ResNet50	66.4	83.4	74.3	87.3	52.5	78.0	70.1	84.2	78.0	80.7	81.3	56.8	63.7	73.3	71.9	71.2	83.4	62.3	55.6	90.0	73.2	8.0
		ResNet101	80.8	**87.7**	80.6	89.8	**57.8**	80.9	75.2	**90.0**	**82.9**	84.5	83.6	63.2	**67.3**	72.6	**79.2**	70.4	90.0	70.7	58.8	**90.3**	77.8	5.9
NPMMR-Det	2020	Darknet53	88.1	87.1	82.0	92.6	49.6	80.6	73.4	83.7	70.5	**85.0**	84.8	64.8	63.3	91.6	71.3	**76.1**	92.0	69.1	58.6	83.6	77.4	34.0
NPMMR-Det*	2020	Darknet53	**89.5**	87.1	**82.7**	**92.7**	51.6	**81.3**	73.1	85.2	71.0	84.3	**85.0**	**65.9**	64.3	**91.7**	74.3	74.4	**92.7**	**73.1**	**60.4**	82.3	**78.1**	11.5

Note: Bold font indicates the best results. The object categories are explained as follows.

C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15	C16	C17	C18	C19	C20
Airplane	Airport	Baseball field	Basketball court	Bridge	Chimney	Dam	Expressway service area	Expressway toll station	Golf course	Ground track field	Harbor	Overpass	Ship	Stadium	Storage tank	Tennis court	Train station	Vehicle	Windmill

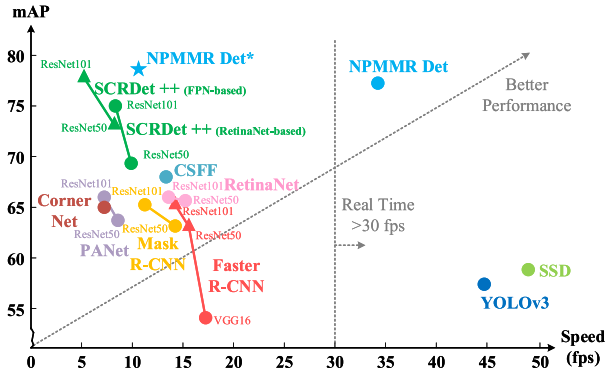

Fig. 12. Comparing the mAP and Speed with State-of-the-art Detectors on the DIOR Data set. Note: In the field of OD, a detector with a speed greater than 30 fps is usually regarded as a real-time detector [8], [56].

case, the inference time of detecting an image is 0.08681s. In the following comparison with the state-of-the-art methods, the best results of NPMMR-Det and NPMMR-Det* are shown.

D. Comparative Experiments and Discussions

To compare the detection performance of NPMMR-Det and state-of-the-art methods, comparative experiments are conducted on commonly used RSOD data sets and visualization results are given. Extensive and state-of-the-art OD methods are selected as the control groups. They not only include methods such as Faster R-CNN [5], SSD [9], YOLOv3 [56], etc. which are widely used and improved in the field of computer vision, but also include SCRDet [14], FMSSD [17], CSFF [59], etc. specifically designed for RSOD. Furthermore, these control groups include both one-stage and two-stage methods. The comparative experiments are fair and extensive.

1) Comparative Experiments and Discussions on the DIOR Data Set: The comparative results on the DIOR data set are shown in Table IX and Fig. 12. Because some articles [54], [58] only publish the mAP of detection (the metric of detection accuracy) without the detection speed, we reimplement them and test the detection speed in the same environment for fair and comprehensive comparisons. For the open-source detectors, the author's source codes are chosen for testing; for some popular detectors like Faster R-CNN, the Detectronv2, and the MMDetection projects are used.

The experimental results show that the mAP of NPMMR-Det reaches 77.4, which is significantly higher than the mAP of the other detectors investigated, except the SCRDet++ (FPN-based) using the Resnet101 [58]. The mAP of NPMMR-Det* using multiscale testing reaches 78.1, which is 0.3 higher than that of the two-stage detector SCRDet++ (FPN-based) using the Resnet101 [58] and is the highest mAP among the state-of-the-art detectors as far as we know. Comparing the detection speed of these detectors, NPMMR-Det reaches 34.0 fps; this is faster than the other detectors are listed in Table IX, except for YOLOv3 [56] and SSD [9]—for those, the mAP is much higher. Correspondingly, in Fig. 12, the detector closest to the upper right corner has a better performance. The proposed NPMMR-Det is excellent both in detection accuracy and detection speed, even if the NPMMR-Det* using multiscale testing can reach a speed of 11.5 fps, which is still faster than the detectors like PANet [38], CornerNet [61], CSFF [59], Mask R-CNN (Resnet101) [60], and SCRDet++ [58].

In Table IX, more detailed experimental results of the state-of-the-art detectors on the DIOR data set are given. The optimal results are shown in bold. The NPMMR-Det*+ obtains the best AP performance in half of the 20 categories, and

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TABLE X
MORE DETAILED RESULTS OF COMPARATIVE EXPERIMENTS ON THE DOTA DATA SET

Methods	Year	Stage	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP	Speed (fps)
Faster R-CNN with FPN [21]	2017	Two-stage	80.32	77.55	32.86	68.13	53.66	52.49	50.04	90.41	75.05	59.59	57.00	49.81	61.69	56.46	41.85	60.46	14.6
ICN [62]	2018	Two-stage	90.00	77.70	53.40	73.30	73.50	65.00	78.20	90.80	79.10	84.80	57.20	62.10	73.50	70.20	58.10	72.50	-
IoU-Adaptive RCNN [63]	2019	Two-stage	88.62	80.22	53.18	66.97	76.30	72.59	84.07	90.66	80.95	76.24	57.12	66.65	84.08	66.36	56.85	72.72	5.0
SCRDet [14]	2019	Two-stage	90.18	81.88	55.30	73.29	72.09	77.65	78.06	90.91	82.44	86.39	64.53	63.45	75.77	78.21	60.11	75.35	5.8
FADet [13]	2019	Two-stage	90.15	78.60	51.92	**75.23**	73.60	71.27	81.41	90.85	83.94	84.77	58.91	65.65	76.92	79.36	68.17	75.38	-
Mask OBB [64]	2019	Two-stage	89.69	**87.07**	58.51	72.04	78.21	71.47	85.20	89.55	84.71	86.76	54.38	70.21	78.98	77.46	**70.40**	76.98	-
A2RMNet [65]	2019	Two-stage	89.84	83.39	60.06	73.46	**79.25**	83.07	87.88	90.90	87.02	87.35	60.74	69.05	79.88	79.74	65.17	78.45	8.0
FMSSD [17]	2019	One-stage	89.11	81.51	48.22	67.94	69.23	73.56	76.87	90.71	82.67	73.33	52.65	67.52	72.37	80.57	60.15	72.43	21.1
EFR [18]	2019	One-stage	88.36	83.90	45.78	67.24	76.80	77.15	85.35	90.77	85.55	75.77	54.64	60.76	71.40	77.90	60.94	73.49	11.0
SCRDet++ (RetinaNet-based) [58]	2020	One-stage	87.89	84.64	56.94	68.03	74.67	78.75	78.50	90.80	85.60	84.98	53.56	56.75	76.66	75.08	62.75	74.37	7.3
SCRDet++ (FPN-based) [58]	2020	Two-stage	90.01	82.32	**61.94**	68.62	69.62	81.17	78.83	90.86	86.32	85.10	65.10	61.12	77.69	**80.68**	64.25	76.24	5.9
NPMMR-Det	2020	One-stage	89.44	83.18	54.50	66.10	76.93	84.08	88.25	90.87	**88.29**	86.32	49.95	68.16	79.61	79.51	57.26	76.16	**34.0**
NPMMR-Det*	2020	One-stage	**93.01**	83.51	61.37	70.77	75.10	**85.14**	**90.49**	**95.30**	86.26	**90.47**	**67.78**	**71.76**	**80.27**	71.41	61.36	**78.93**	11.5

Note: Bold font indicates the best results. Since some methods' codes are not open-source, we cannot test their detection speed, which is indicated by "-". The object categories are explained as follows.

PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC
Plane	Baseball diamond	Bridge	Ground field track	Small vehicle	Large vehicle	Ship	Tennis court	Basketball court	Storage tank	Soccer-ball field	Roundabout	Harbor	Swimming pool	Helicopter

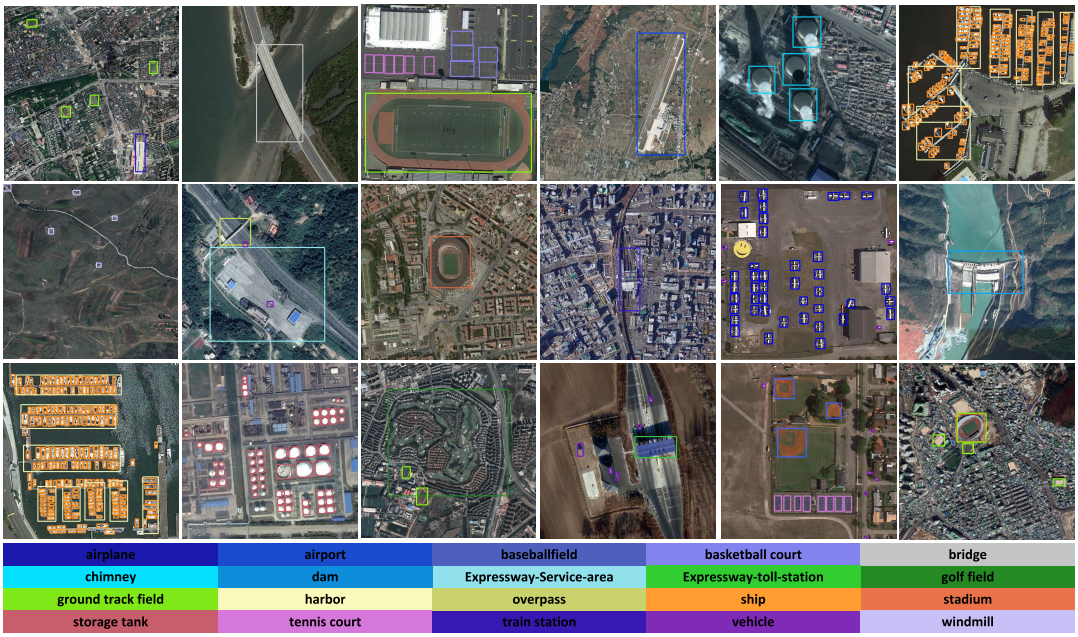

Fig. 13. Visualization Results of the proposed NPMMR-Det on the DIOR Data set.

its mAP is also the best of the articles we have investigated. In summary, the comparative experiments on the DIOR data set to confirm that NPMMR-Det has state-of-the-art performance compared with the other existing detectors. To illustrate the detection performance more intuitively, Fig. 13 shows the visual results of the proposed NPMMR-Det on the DIOR data set.

2) Comparative Experiments and Discussions on the DOTA Data Set: In order to further verify the effectiveness and versatility of NPMMR-Det in the RSOD task, comparative experiments on the DOTA data set are conducted.

As shown in Table X, NPMMR-Det has the highest mAP, 76.16, on the DOTA data set among all the one-stage detectors, like FMSSD [17], EFR [18], and SCRDet++ (RetinaNet-based) [58]. This result is also better than the performance of some two-stage detectors like Faster R-CNN [5], ICN [62], IoU-Adaptive RCNN [63], SCRDet [14], and FADet [13]. As is generally known, the mAP of a two-stage detector is usually higher than that of a one-stage detector, but the two-stage detector's speed is very slow. NPMMR-Det keeps the detection speed at 34.0 fps with the mAP of 76.16. The detection speed of NPMMR-Det is the fastest compared with the state-of-the-art detectors are listed in Table X. In summary, these results not only demonstrate the better performance of NPMMR-Det but also reflect its practical application value and potential.

For the high mAP, NPMMR-Det* adopts multiscale testing and gets an mAP of 78.93. This result exceeds the mAPs of the two-stage detectors like Mask OBB [64], A2RMNet [65], and SCRDet++ (FPN-based) [58]. In this case, the speed of our

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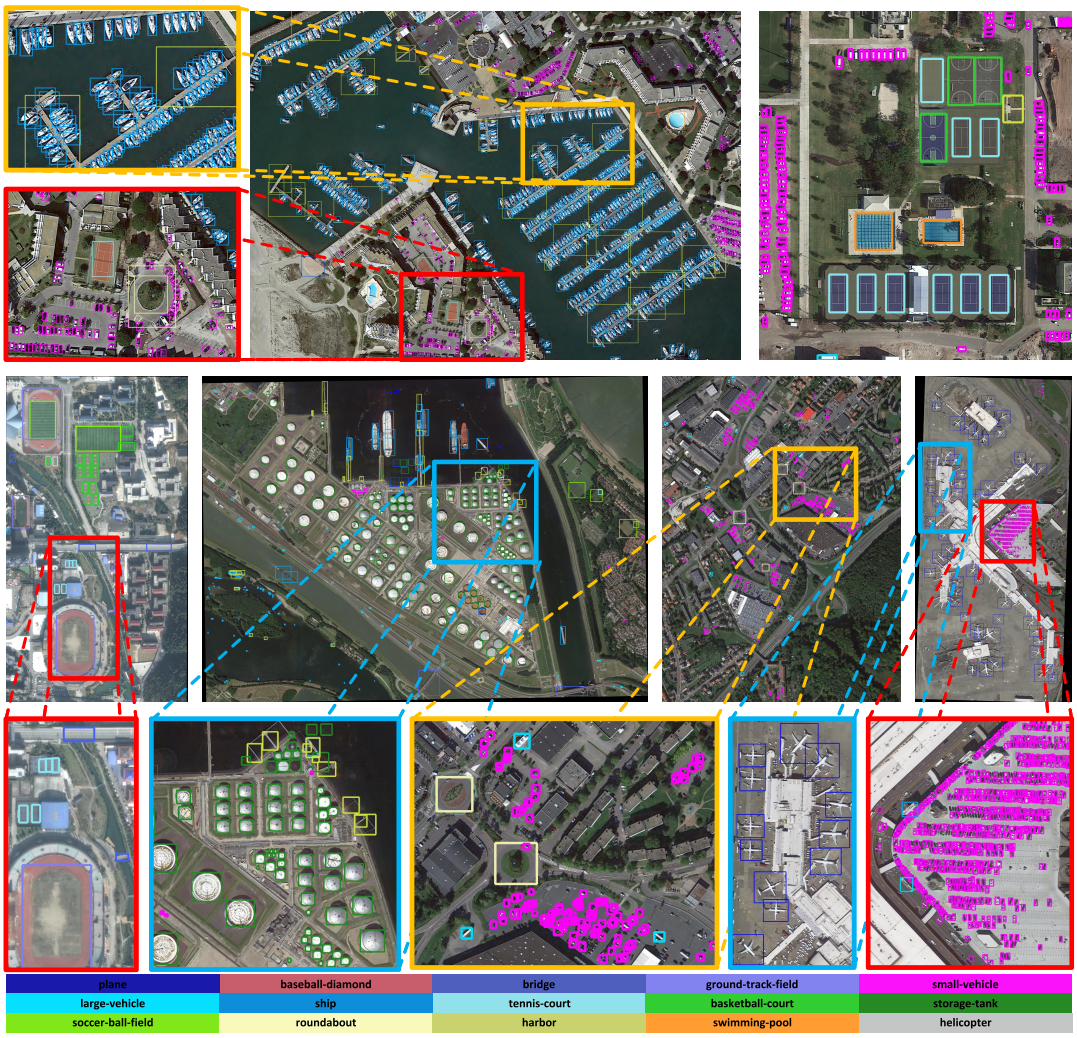

Fig. 14. Visualization Results of the proposed NPMMR-Det on the DOTA Data set.

NPMMR-Det* is 11.5 fps, which is faster than the detectors compared above.

In addition, to illustrate the detection performance of NPMMR-Det more intuitively, Fig. 14 shows the visualization results on the DOTA data set. Since the image size in the DOTA data set is very large, it shows the detection results on the whole images and on the subimages that are obtained by enlarging the partial detection results on the whole images. The results reflect satisfying detection performance further ensure the effective of the proposed NP-MMR-Det.

V. Conclusion

In this article, a novel detector NPMMR-Det is proposed for refining the misalignments of the RSOD task. In NPMMR-Det, NP-Attention is designed to solve the misalignment between the network's focus and the objects. The MSR-FPN is developed to refine the misalignment between the respective scales of the features and the objects. Finally, the MTR-Head is raised to alleviate the feature misalignment between localization and classification.

The experiments have demonstrated the following: 1) NP-Attention makes the network focus more on the valuable object features efficiently with high performance; 2) the MSR-FPN fuses the multiscale context features and adapts the receptive field better to the scale of objects, improving the multiscale detection accuracy; and 3) the MTR-Head enables the different task branches of the network to learn more suitable features and dynamically adjusts the localization loss based on the object's scale. Finally, the proposed NPMMR-Det achieves state-of-the-art performance on public data sets.

Despite its demonstrated benefits, this work could be further improved. In the future, we will focus primarily on improving the MTR-Head for oriented box detection, finding more robust multiscale and multidirectional features, and designing a faster and lighter detector, NPMMR-Det-Lite, for the embedded devices.

Our source code and pretrained models have been available in *https://github.com/Shank2358* to ensure that the results of this article can be reproduced.

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REFERENCES

[1] P. F. Felzenszwalb, R. B. Girshick, D. McAllester, and D. Ramanan, "Object detection with discriminatively trained part-based models," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 32, no. 9, pp. 1627–1645, Sep. 2010.

[2] H.-Y. Cheng, C.-C. Weng, and Y. Chen, "Vehicle detection in aerial surveillance using dynamic Bayesian networks," *IEEE Trans. Image Process.*, vol. 21, no. 4, pp. 2152–2159, Mar. 2012.

[3] Z. Chen *et al.*, "Vehicle detection in high-resolution aerial images via sparse representation and superpixels," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 103–116, Jan. 2016.

[4] J. Han, D. Zhang, G. Cheng, N. Liu, and D. Xu, "Advanced deep-learning techniques for salient and category-specific object detection: A survey," *IEEE Signal Process. Mag.*, vol. 35, no. 1, pp. 84–100, Jan. 2018.

[5] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.

[6] J. Dai, Y. Li, K. He, and J. Sun, "R-FCN: Object detection via region-based fully convolutional networks," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, Barcelona, Spain, Dec. 2016, pp. 379–387.

[7] Z. Cai and N. Vasconcelos, "Cascade R-CNN: Delving into high quality object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Salt Lake City, UT, USA, Jun. 2018, pp. 6154–6162.

[8] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Las Vegas, NV, USA, Jun. 2016, pp. 779–788.

[9] W. Liu *et al.*, "SSD: Single shot multibox detector," in *Proc. Eur. Conf. Comput. Vis.*, Amsterdam, The Netherlands, Oct. 2016, pp. 21–37.

[10] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis.*, Venice, Italy, Oct. 2017, pp. 2999–3007.

[11] G. Cheng, P. Zhou, and J. Han, "Learning rotation-invariant convolutional neural networks for object detection in VHR optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 7405–7415, Dec. 2016.

[12] G. Cheng, J. Han, P. Zhou, and D. Xu, "Learning rotation-invariant and Fisher discriminative convolutional neural networks for object detection," *IEEE Trans. Image Process.*, vol. 28, no. 1, pp. 265–278, Jan. 2019.

[13] C. Li, C. Xu, Z. Cui, D. Wang, T. Zhang, and J. Yang, "Feature-attentioned object detection in remote sensing imagery," in *Proc. IEEE Int. Conf. Image Process.*, Taipei, China, Sep. 2019, pp. 3886–3890.

[14] X. Yang *et al.*, "SCRDet: Towards more robust detection for small, cluttered and rotated objects," in *Proc. IEEE Int. Conf. Comput. Vis.*, Seoul, South Korea, Oct. 2019, pp. 8232–8241.

[15] G. Zhang, S. Lu, and W. Zhang, "CAD-Net: A context-aware detection network for objects in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 10015–10024, Aug. 2019.

[16] P. Sun, G. Chen, G. Luke, and Y. Shang, "Salience biased loss for object detection in aerial images," 2018, *arXiv:1810.08103*. [Online]. Available: http://arxiv.org/abs/1810.08103

[17] P. Wang, X. Sun, W. Diao, and K. Fu, "FMSSD: Feature-merged single-shot detection for multiscale objects in large-scale remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3377–3390, Dec. 2020.

[18] K. Fu, Z. Chen, Y. Zhang, and X. Sun, "Enhanced feature representation in detection for optical remote sensing images," *Remote Sens.*, vol. 11, no. 18, p. 2095, 2019.

[19] C. Wang, X. Bai, S. Wang, J. Zhou, and P. Ren, "Multiscale visual attention networks for object detection in VHR remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 2, pp. 310–314, Feb. 2019.

[20] X. Wang, R. Girshick, A. Gupta, and K. He, "Non-local neural networks," in *Proc. IEEE Int. Conf. Comput. Vis.*, Salt Lake City, UT, USA, Jun. 2018, pp. 7794–7803.

[21] T.-Y. Lin, P. Dollar, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Honolulu, HI, USA, Jul. 2017, pp. 936–944.

[22] K. Fu, Z. Chang, Y. Zhang, G. Xu, K. Zhang, and X. Sun, "Rotation-aware and multi-scale convolutional neural network for object detection in remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 161, pp. 294–308, 2020.

[23] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking atrous convolution for semantic image segmentation," 2017, *arXiv:1706.05587*. [Online]. Available: http://arxiv.org/abs/1706.05587

[24] Y. Wu *et al.*, "Rethinking classification and localization for object detection," in *Proc. IEEE Comput. Vis. Pattern Recognit.*, Jun. 2020, pp. 10186–10195.

[25] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Salt Lake City, UT, USA, Jun. 2018, pp. 7132–7141.

[26] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional Block Attention Module," in *Proc. Eur. Conf. Comput. Vis.*, Munich, Germany, Sep. 2018, pp. 3–19.

[27] Y. Cao, J. Xu, S. Lin, F. Wei, and H. Hu, "GCNet: Non-local networks meet squeeze-excitation networks and beyond," in *Proc. IEEE Int. Conf. Comput. Vis. Workshop*, Seoul, South Korea, Oct. 2019, pp. 1971–1980.

[28] Y. Yuan and J. Wang, "OCNet: Object context network for scene parsing," 2018, *arXiv:1809.00916*. [Online]. Available: http://arxiv.org/abs/1809.00916

[29] K. Yue, M. Sun, Y. Yuan, F. Zhou, E. Ding, and F. Xu, "Compact generalized non-local network," in *Proc. Annu. Conf. Neural Inf. Process. Syst.*, Montréal, QC, Canada, Dec. 2018, pp. 6510–6519.

[30] J. Fu *et al.*, "Dual attention network for scene segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Long Beach, CA, USA, Jun. 2019, pp. 3146–3154.

[31] Z. Huang, X. Wang, L. Huang, C. Huang, Y. Wei, and W. Liu, "CCNet: Criss-cross attention for semantic segmentation," in *Proc. IEEE Int. Conf. Comput. Vis.*, Seoul, South Korea, Oct. 2019, pp. 603–612.

[32] Z. Zhu, M. Xu, S. Bai, T. Huang, and X. Bai, "Asymmetric non-local neural networks for semantic segmentation," in *Proc. IEEE Int. Conf. Comput. Vis.*, Seoul, South Korea, Oct. 2019, pp. 593–602.

[33] P. Sermanet, D. Eigen, X. Zhang, M. Mathieu, R. Fergus, and Y. LeCun, "OverFeat: Integrated recognition, localization and detection using convolutional networks," in *Proc. Int. Conf. Learn. Represent.*, Banff, AB, Canada, Apr. 2014, pp. 1–16.

[34] B. Singh and L. S. Davis, "An analysis of scale invariance in object detection—SNIP," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Salt Lake City, UT, USA, Jun. 2018, pp. 3578–3587.

[35] Z. Cai, Q. Fan, R. S. Feris, and N. Vasconcelos, "A unified multi-scale deep convolutional neural network for fast object detection," in *Proc. 14th Eur. Conf. Comput. Vis.*, Amsterdam, The Netherlands, Oct. 2016, pp. 354–370.

[36] F. Yang, W. Choi, and Y. Lin, "Exploit all the layers: Fast and accurate CNN object detector with scale dependent pooling and cascaded rejection classifiers," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Las Vegas, NV, USA, Jun. 2016, pp. 2129–2137.

[37] J. Li, X. Liang, S. Shen, T. Xu, J. Feng, and S. Yan, "Scale-aware fast R-CNN for pedestrian detection," *IEEE Trans. Multimedia*, vol. 20, no. 4, pp. 985–996, Oct. 2018.

[38] K. Wang, J. H. Liew, Y. Zou, D. Zhou, and J. Feng, "PANet: Few-shot image semantic segmentation with prototype alignment," in *Proc. IEEE Int. Conf. Comput. Vis.*, Seoul, South Korea, Oct. 2019, pp. 9197–9206.

[39] M. Tan, R. Pang, and Q. V. Le, "EfficientDet: Scalable and efficient object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2020, pp. 10781–10790.

[40] T. Liang, Y. Wang, Q. Zhao, H. Zhang, Z. Tang, and H. Ling, "MFPN: A novel mixture feature pyramid network of multiple architectures for object detection," 2019, *arXiv:1912.09748*. [Online]. Available: http://arxiv.org/abs/1912.09748

[41] Y. Li, Y. Chen, N. Wang, and Z.-X. Zhang, "Scale-aware trident networks for object detection," in *Proc. IEEE Int. Conf. Comput. Vis.*, Seoul, South Korea, Oct. 2019, pp. 6054–6063.

[42] Z. Li, C. Peng, G. Yu, X. Zhang, Y. Deng, and J. Sun, "Light-head R-CNN: In defense of two-stage object detector," 2017, *arXiv:1711.07264*. [Online]. Available: http://arxiv.org/abs/1711.07264

[43] A. Li, X. Yang, and C. Zhang, "Rethinking classification and localization for cascade R-CNN," in *Proc. British Mach. Vis. Conf.*, Cardiff, U.K., Sep. 2019, p. 241.

[44] B. Jiang, R. Luo, J. Mao, T. Xiao, and Y. Jiang, "Acquisition of localization confidence for accurate object detection," in *Proc. Eur. Conf. Comput. Vis.*, Munich, Germany, Sep. 2018, pp. 816–832.

[45] H. Zhang and J. Wang, "Towards adversarially robust object detection," in *Proc. IEEE Int. Conf. Comput. Vis.*, Seoul, South Korea, Oct. 2019, pp. 421–430.

[46] Z. Yang, S. Liu, H. Hu, L. Wang, and S. Lin, "RepPoints: Point set representation for object detection," in *Proc. IEEE Int. Conf. Comput. Vis.*, Seoul, South Korea, Oct. 2019, pp. 9657–9666.

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[47] Y. He, C. Zhu, J. Wang, M. Savvides, and X. Zhang, "Bounding box regression with uncertainty for accurate object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Long Beach, CA, USA, Jun. 2019, pp. 2888–2897.
[48] H. Rezatofighi, N. Tsoi, J. Gwak, A. Sadeghian, I. Reid, and S. Savarese, "Generalized intersection over union: A metric and a loss for bounding box regression," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Los Angeles, CA, USA, Jun. 2019, pp. 658–666.
[49] Z. Zheng, P. Wang, W. Liu, J. Li, R. Ye, and D. Ren, "Distance-IoU loss: Faster and better learning for bounding box regression," in *Proc. AAAI Conf. Artif. Intell.*, New York, NY, USA, vol. 34, Feb. 2020, pp. 12993–13000.
[50] Y. Wu and K. He, "Group normalization," 2018, *arXiv:1803.08494*. [Online]. Available: http://arxiv.org/abs/1803.08494
[51] R. Zhang, "Making convolutional networks shift-invariant again," in *Proc. Int. Conf. Mach. Learn.*, Long Beach, CA, USA, Jun. 2019, pp. 7324–7334.
[52] J. Dai *et al.*, "Deformable convolutional networks," in *Proc. IEEE Int. Conf. Comput. Vis.*, Venice, Italy, Oct. 2017, pp. 764–773.
[53] X. Zhu, H. Hu, S. Lin, and J. Dai, "Deformable convnets V2: More deformable, better results," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Long Beach, CA, USA, Jun. 2019, pp. 9308–9316.
[54] K. Li, G. Wan, G. Cheng, L. Meng, and J. Han, "Object detection in optical remote sensing images: A survey and a new benchmark," *ISPRS J. Photogramm. Remote Sens.*, vol. 159, pp. 296–307, 2020.
[55] G.-S. Xia *et al.*, "DOTA: A large-scale dataset for object detection in aerial images," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Salt Lake City, UT, USA, Feb. 2018, pp. 3974–3983.
[56] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," 2018, *arXiv:1804.02767*. [Online]. Available: http://arxiv.org/abs/1804.02767
[57] Z. Zhang, T. He, H. Zhang, Z. Zhang, J. Xie, and M. Li, "Bag of freebies for training object detection neural networks," 2019, *arXiv:1902.04103*. [Online]. Available: http://arxiv.org/abs/1902.04103
[58] X. Yang, J. Yan, X. Yang, J. Tang, W. Liao, and T. He, "SCRDet++: Detecting small, cluttered and rotated objects via instance-level feature denoising and rotation loss smoothing," 2020, *arXiv:2004.13316*. [Online]. Available: http://arxiv.org/abs/2004.13316
[59] G. Cheng, Y. Si, H. Hong, X. Yao, and L. Guo, "Cross-scale feature fusion for object detection in optical remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, early access, Mar. 4, 2020, doi: 10.1109/LGRS.2020.2975541.
[60] K. He, G. Gkioxari, P. Dollar, and R. Girshick, "Mask R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis.*, Venice, Italy, Oct. 2017, pp. 2980–2988.
[61] H. Law and J. Deng, "CornerNet: Detecting objects as paired keypoints," in *Proc. Eur. Conf. Comput. Vis.*, Munich, Germany, Jun. 2018, pp. 734–750.
[62] S. M. Azimi, E. Vig, R. Bahmanyar, M. Körner, and P. Reinartz, "Towards multi-class object detection in unconstrained remote sensing imagery," in *Proc. Asian Conf. Comput. Vis.*, Perth, WA, Australia, Dec. 2018, pp. 150–165.
[63] J. Yan, H. Wang, M. Yan, D. Wenhui, X. Sun, and H. Li, "IoU-adaptive deformable R-CNN: Make full use of IoU for multi-class object detection in remote sensing imagery," *Remote Sens.*, vol. 11, no. 3, p. 286, Feb. 2019.
[64] J. Wang, J. Ding, H. Guo, W. Cheng, T. Pan, and W. Yang, "Mask OBB: A semantic attention-based mask oriented bounding box representation for multi-category object detection in aerial images," *Remote Sens.*, vol. 11, no. 24, p. 2930, 2019.
[65] H. Qiu, H. Li, Q. Wu, F. Meng, K. N. Ngan, and H. Shi, "A2RMNet: Adaptively aspect ratio multi-scale network for object detection in remote sensing images," *Remote Sens.*, vol. 11, no. 13, p. 1594, 2019.

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